

Experimental determination of Triple Higgs Couplings in the Real singlet extension of the Standard Model at the HL-LHC

(Determinación experimental de las acoplamientos triples del Higgs
en la extensión real singlete del Modelo Estándar en el HL-LHC)

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Abstract

In this work, we analyze the real singlet extension of the Standard Model (RxSM), focusing on the parameter space that yields a strong first-order electroweak phase transition (SFOEWPT) in the early universe. This scenario is of particular interest as it allows that electroweak baryogenesis could explain the observed matter-antimatter asymmetry. It predicts two Higgs bosons: h , which we assume to be the Higgs boson discovered at the LHC, H , a heavier second $\mathcal{CP}$ -even Higgs.

Our study evaluates di-Higgs production channels (hh , hH , and HH) via gluon fusion at the High-Luminosity LHC (HL-LHC) with a center-of-mass energy of $\sqrt{s} = 14$ TeV. We investigate the impact of the triple Higgs couplings (THCs): λ_{hhh} , λ_{hhH} , λ_{hHH} , and λ_{HHH} , and consider the potential enhancement of double light Higgs production driven by resonant s -channel diagrams involving the heavy Higgs boson, H , at tree level.

We first explore the benchmark plane where all points are compatible with a SFOEWPT. Using the tool **anyBSM**, we compute the expected cross sections and expected significances for all di-Higgs final states within the RxSM. Our results indicate that the number of expected events for channels containing at least one BSM Higgs boson in the final state (hH and HH) is statistically insignificant at the HL-LHC. Consequently, only the double light Higgs production channel (hh) offers viable experimental sensitivity. Finally, focusing on this channel, we perform a χ^2 fit based on sensitive kinematic observables to extract the projected experimental constraints on the relevant trilinear Higgs couplings. We find that for specific benchmark points two regions appeared for $\lambda_{hhH} < 0$ and $\lambda_{hhH} > 0$, but for the rest of the benchmark points only one contour appeared, concluding that depending which BP is realized in nature, a first determination of λ_{hhH} might be possible.



Resumen

En este trabajo se ha analizado la extensión de singlete del Modelo Estándar (del inglés RxSM), centrándose en el espacio de parámetros que produzcan una transición de fase fuerte de primer orden (del inglés SFOEWPT) en el universo primitivo. Este escenario es de gran importancia debido a que permite la bariogénesis electro-débil puede explicar la asimetría entre la materia y antimateria. El modelo predice dos bosones de Higgs: h , el cual se comprende como el boson de Higgs descubierto en el LHC, y H , un segundo Higgs más pesado con paridad $\mathcal{CP}$ par.

En el estudio realizado se analizan los canales de producción de dos Higgses (hh , hH y HH) a través de la fusión de gluones en el *High-Luminosity* LHC (HL-LHC) con una energía de centro de masas de $\sqrt{s} = 14$ TeV. Se han investigado el impacto de los acoplamientos triple Higgs (del inglés THCs): λ_{hhh} , λ_{hhH} , λ_{hHH} y λ_{HHH} , y considerado la mejora potencial de la producción de dos bosones Higgs generada por la resonancia en el diagrama de canal s donde está involucrado el bosón Higgs pesado a nivel de árbol.

Primeramente se exploró el plano de referencia, donde todos los puntos presentes son compatibles con una SFOEWPT. Utilizando la herramienta de **anyBSM**, se calcularon las secciones eficaces totales esperadas y las significancias logradas para todas las desintegraciones a dos Higgs admitidas dentro del modelo RxSM. Los resultados muestran que los procesos donde se crea al menos un Higgs pesado (hH y HH) son estadísticamente insignificantes dentro del HL-LHC. Por consiguiente, únicamente el canal de producción de dos bosones de Higgs ligeros (hh) ofrece una sensibilidad experimental viable. Finalmente, centrándonos en este último canal, se realizó un ajuste χ^2 basado en observables cinemáticos para extraer los límites de los acoplamientos triple Higgs relevantes. Encontramos que, para puntos de referencia específicos, aparecieron dos regiones para $\lambda_{hhH} < 0$ y $\lambda_{hhH} > 0$, pero para el resto de los puntos de referencia solo apareció un contorno; concluyendo así que, dependiendo de qué BP se realice en la naturaleza, podría ser posible una primera determinación de λ_{hhH} .



Index

1	Introduction	1
1.1	Strong First Order Electroweak Phase Transition	2
1.2	HL-LHC and di-Higgs production	3
2	The model and the Constrains	4
2.1	General Singlet Extension of the SM (RxSM)	4
2.2	Theoretical and Experimental Constraints	8
2.3	Alignment limit	9
2.3.1	Alignment without decoupling	9
2.3.2	Alignment with decoupling	10
2.3.3	Benchmark Plane	11
3	Di-Higgs Production in the HL-LHC	16
3.1	anyBSM: anyHH and anyH3	17
3.2	Parton Distribution Functions	17
3.3	Branching Ratios	18
3.4	Total cross section and Number of expected events	20
4	Results	22
4.1	Total Cross Section	22
4.1.1	Parametric Impact on Di-Higgs Production	23
4.1.2	Expected Signal Yields	24
4.2	Extraction of Triple Higgs Couplings	25

5	Conclusions	33
A	Appendix	35
A.1	Total cross section expressions for two scalar Higgs bosons $g^a g^b \rightarrow H_i H_j$	35
A.1.1	Double light Higgs boson production ($gg \rightarrow hh$)	36
A.2	Matrix element behavior	38
	Referencias	45

In the last century, the huge development of quantum physics has led to astonishing discoveries. All this progress has driven to the formulation of the Standard Model (SM), a highly successful theoretical framework, which is recognized as the most successful theory in particle physics, validated by decades of experimental successes. This theory has accomplished to describe many of the experimental data such as the measurements of W and Z bosons in the 1983 [[1], [2], [3], [4]]. The most recent prediction of this theory was the Higgs boson, predicted in the 1964 [[5], [6]] and discovered in the 2012 at the LHC [[7], [8]]. Despite being that accurate, this theory can not explain certain opened problems in physics: Gravity, dark matter and energy or even the neutrino mass. One of those unexplained problems is the asymmetry between the matter and antimatter in the universe, which the SM fails to give an explanation for, therefore, Beyond Standard Model (BSM) theories are required to solve this inconsistency.

One of the mechanism that are used for explaining the matter-antimatter asymmetry is the electroweak baryogenesis (EWBG), which refers to a mechanism that produces an asymmetry in the density of baryons during the electroweak phase transition (EWPT) [9]. For a model to successfully generate baryogenesis, it must satisfy the Sakharov conditions: violation in the baryonic number, violation of $\mathcal{C}$ and $\mathcal{CP}$ symmetry and violation of thermal equilibrium [10]. The SM only satisfies completely the first condition via electroweak sphaleron transitions. Regarding the second condition, while $\mathcal{CP}$ violation is successfully incorporated into the SM and experimentally confirmed in neutral Kaon decays, as mentioned in [11], it is quantitatively insufficient. These $\mathcal{CP}$ -violating effects originating from the CKM matrix are several orders of magnitude smaller than what is required to explain the observed cosmological baryon symmetry [9]. Finally, the third Sakharov condition fails to be satisfied within the SM framework. For the EWPT to be first-order, the SM Higgs boson mass should need to be $m_h \lesssim 80$ GeV [9]. Consequently, the existence of beyond the SM (BSM) physics coupled to the SM with sufficient strength to drive Strong First Order EWPT (SFOEWPT) is necessary to understand the matter anti-matter asymmetry problem.

Throughout this thesis, the real singlet extension of the SM (RxSM) is proposed as a viable

theoretical framework to fulfill the third Sakharov condition through a SFOEWPT that allows EWBG. The main objective is to explore this model in order to reconstruct the extended Higgs potential. For this purpose, specific benchmark points (BPs) known to drive a SFOEWPT [[12]] are selected and their corresponding Trilinear Higgs Couplings (THCs) are analyzed. To assess the experimental accessibility of these scenarios, all available di-Higgs production channels in the RxSM model are considered, computing their total cross sections within the upcoming high luminosity phase of the Large Hadron Collider (LHC), which has not yet been studied in the most recent researches. With the use of the public code `anyBSM`, the entire numerical evaluation was computed.

1.1 Strong First Order Electroweak Phase Transition

As mentioned before, the discovery of the Higgs boson in 2012 generated a big interesting about the nature and the properties of the mechanism which provides particles to have mass in the SM. This phenomenon is driven by Electroweak Symmetry Breaking (EWSB) and in order to understand this process deeper one has to go to the early universe, when the EWPT occurred. In the radiation-dominated age Universe the full $SU(2)_L \times U(1)_Y$ electroweak symmetry was present [9], but as the temperature of the early Universe dropped below the electroweak phase transition scale ($T_c \sim 160$ GeV) [13], the Higgs field acquired a non-zero vacuum expectation value v . This spontaneously broke the $SU(2)_L \times U(1)_Y$ electroweak gauge symmetry down to electromagnetic symmetry $U(1)$.

The Higgs mechanism is known to be temperature-dependent, meaning that thermal effects introduce corrections to the scalar potential. Nowadays, at $T = 0$, the minimum of the potential breaks the electroweak gauge symmetry. However, cosmology tells us that in the early universe this symmetry was not broken and the minimum of the potential was set at $v = 0$, this means that the minimum of the potential must have changed from zero to a non-zero values. This transition is what is know as EWPT.

As in thermodynamics, there are two types of phase transitions: first order phase transition and second order phase transition. In Fig. 1.1, the left side shows the potential for a first-order phase transition, while the right side displays it for a second-order phase transition. Second order phase transitions describes continuous changes in the potential along the phase change, this is what is happening in the SM. Nevertheless, in order to preserve the third condition of Sakharov, that is why the EWPT must be driven by a First Order EWPT (FOEWPT) [9], so the Higgs potential needs a cubic term to achieve this type of transition. The SM alone can not provide such a term with the measured Higgs mass of 125 GeV. This fundamental weakness motivates BSM frameworks like the RxSM, where the interactions between the SM doublet and the new scalar singlet dynamically supply the missing cubic-like terms required to drive the first-order phase transition.

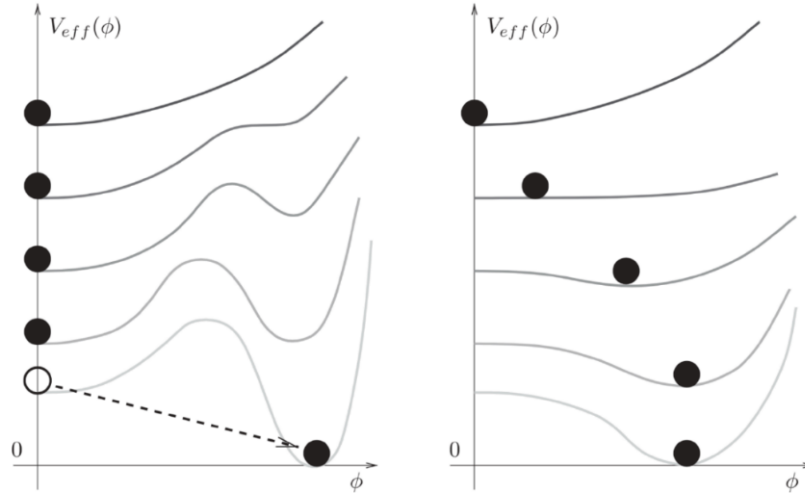


Figure 1.1: The evolution of the potential as a function of temperature, the darker curve is the higher temperature and the lighter curve at $T = 0$. The black dot represents the evolution of the VEV. Left: First Order Transition. Right: Second Order Transition. Image extracted from [14].

1.2 HL-LHC and di-Higgs production

The High Luminosity Large Hadronic Collider (HL-LHC) is an upcoming experiment in the 2030, which is part of the Large Hadronic Collider (LHC) located at CERN, Switzerland. The LHC is a circular hadronic collider of 27 km which has many different detectors, one of each specialized in one objective. The main four detectors are: CMS, ATLAS, LHCb and ALICE, but apart from them, there are also smaller detectors such as LHCf. The first two main detectors, CMS and ATLAS, are the ones who published the discovery of the Higgs boson in the 2012 [[8], [7]].

HL-LHC is the high luminosity phase of the LHC that will accumulate about ten times more data than all previous LHC runs combined. This new era of the LHC is expected to start gathering data from 2030 until 2040 and it aims to increase the center-of-mass energy to $\sqrt{s} = 14$ TeV and the integrated luminosity to $\mathcal{L} = 3 \text{ ab}^{-1}$ per detector. This growth in the luminosity compared to previous stages of LHC will provide a significant improvement in the possible compiled data which can lead to observe events with low cross sections and enhanced the analysis of them.

This new phase will be very interesting specially for processes like the one which is being studied through this thesis: production of di-Higgs bosons via proton-proton collisions ($pp \rightarrow HH$, where H denotes the Higgs boson discovered in 2012 in the LHC).

Although the double Higgs production in the SM has a tiny cross section compared to the background, recent projections by ATLAS and CMS collaborations demonstrated that the data collected in the upcoming high luminosity phase will allow for the observation of this process with and expected statistical significance exceeding 7σ [15]. This unprecedented precision will be crucial to measure the Higgs trilinear self-coupling and also to verify BSM Higgs sectors that allows the SFOEWPT.

2

The model and the Constrains

2.1 General Singlet Extension of the SM (RxSM)

In this work, a simple BSM model, the most general singlet extension of the SM (RxSM) will be used. This model adds a real singlet S to the Higgs sector of the SM and it does not present $\mathbb{Z}_2$ symmetry of the Lagrangian, as described in [16] and in [17]. After the EWSB, the SM-like doublet Φ and the singlet S can be expanded as

$$\Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{2} \cdot G^+ \\ v + \phi + iG^0 \end{pmatrix} \quad \text{and} \quad S = s + v_s , \quad (2.1)$$

where s is the new scalar field and v_s is the singlet vacuum expectation value (VEV), while ϕ is the $\mathcal{CP}$ -even component of the doublet, G^0 ($G^\pm$) is the neutral (charged) would-be Goldstone bosons¹, and v is the EW (or SM) VEV.

To eliminate the unphysical Goldstone bosons from the Lagrangian and make the physical spectrum manifest, we exploit the local gauge symmetry by going to the unitary gauge. By adopting this gauge, the doublet and the singlet simplify to:

$$\Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + \phi \end{pmatrix} \quad \text{and} \quad S = s + v_s , \quad (2.2)$$

where the degrees of freedom from the Goldstone bosons have been absorbed to provide the longitudinal polarization states of the massive $W^\pm$ and Z bosons. The scalar potential of the RxSM at tree level is written as follows,

¹Goldstone bosons are massless scalar fields arising from spontaneous symmetry breaking. In the SM, these degrees of freedom are absorbed to become the longitudinal polarization states of the massive $W^\pm$ and Z bosons [18].

$$V(\Phi, S) = \mu^2 |\Phi|^2 + \frac{\lambda_H}{2} |\Phi|^4 + \frac{m_S^2}{2} S^2 + \frac{\kappa_S}{3} S^3 + \frac{\lambda_S}{2} S^4 + \kappa_{SH} S |\Phi|^2 + \frac{\lambda_{SH}}{2} S^2 |\Phi|^2 \quad (2.3)$$

After introducing the new Higgs field and not imposing the $\mathbb{Z}_2$ symmetry, the new potential would include linear and cubic terms, leading to seven Lagrangian parameters. However, if the VEV values of the doublet and the singlet fields are considered, the model remains with nine free parameters. These nine parameters can be reduced using the tree-level tadpole equations [19]:

$$0 = \frac{\partial V}{\partial \phi} \Big|_{\phi=0, s=0} = \mu^2 v + \frac{\lambda_H v^3}{2} + \kappa_{SH} v v_S + \frac{\lambda_{SH} v_S^2 v}{2} , \quad (2.4)$$

$$0 = \frac{\partial V}{\partial s} \Big|_{\phi=0, s=0} = m_S^2 v_S + \kappa_S v_S^2 + 2\lambda_S v_S^3 + \frac{\kappa_{SH} v^2}{2} + \frac{\lambda_{SH} v v_S^2}{2} . \quad (2.5)$$

From these conditions it is possible to get the relation for these two Lagrangian parameters μ and m_S :

$$\mu^2 = - \left(\frac{\lambda_H v^2}{2} + \kappa_{SH} v v_S + \frac{\lambda_{SH} v_S^2}{2} \right) , \quad (2.6)$$

$$m_S^2 = - \left(\kappa_S v_S + 2\lambda_S v_S^2 + \frac{\kappa_{SH} v^2}{4v_S} + \frac{\lambda_{SH} v^2}{2} \right) . \quad (2.7)$$

After introducing the unitary gauge expressions of the doublet and singlet from Eq. 2.2 into the potential (Eq. 2.3), the mass matrix $\mathcal{M}$ can be expressed as

$$\mathcal{M}^2 = \left(\begin{array}{cc} \partial_{\phi\phi} V & \partial_{s\phi} V \\ \partial_{\phi s} V & \partial_{ss} V \end{array} \right) \Big|_{s=0, \phi=0} = \left(\begin{array}{cc} \mathcal{M}_{\phi\phi}^2 & \mathcal{M}_{\phi s}^2 \\ \mathcal{M}_{\phi s}^2 & \mathcal{M}_s^2 \end{array} \right) = \left(\begin{array}{cc} v^2 \lambda_H & v(\kappa_{SH} + \lambda_{SH} v_S) \\ v(\kappa_{SH} + \lambda_{SH} v_S) & v_S(4\lambda_S v_S + \kappa_S) - \frac{\kappa_S v^2}{2v_S} \end{array} \right) \quad (2.8)$$

where $\partial_{ij} V = \frac{\partial^2 V}{\partial \phi_i \partial \phi_j}$. In this model, the fields ϕ and s mix to $\mathcal{CP}$ -even eigenstates h and H with masses $m_h < m_H$:

$$\begin{pmatrix} h \\ H \end{pmatrix} = \begin{pmatrix} c_\alpha & s_\alpha \\ -s_\alpha & c_\alpha \end{pmatrix} \begin{pmatrix} \phi \\ s \end{pmatrix} , \quad (2.9)$$

where $c_\alpha \equiv \cos \alpha$ and $s_\alpha \equiv \sin \alpha$ and h and H will be the SM-like Higgs boson and H the new heavy Higgs boson, respectively. Now that the states are related with the fields of the potential, the mass of each of the physical particles are obtained from the mass matrix (Eq. 2.8), which is diagonalized using the rotation matrix $R(\alpha)$:

$$\begin{pmatrix} m_h^2 & 0 \\ 0 & m_H^2 \end{pmatrix} = R^{-1}(\alpha) \mathcal{M}^2 R(\alpha) . \quad (2.10)$$

Solving this eigenvalue problem, the eigenvalues of these equation will be the associated masses of both Higgs bosons:

$$m_{h,H}^2 = \frac{1}{2} \left(\mathcal{M}_\phi^2 + \mathcal{M}_s^2 \mp |\mathcal{M}_\phi^2 - \mathcal{M}_s^2| \sqrt{1 + \frac{2\mathcal{M}_{\phi s}^2}{\mathcal{M}_\phi^2 - \mathcal{M}_s^2}} \right) . \quad (2.11)$$

From the diagonalization of the mass matrix, the mixing angle can also be computed as

$$\tan 2\alpha = \frac{2\mathcal{M}_{\phi s}^2}{\mathcal{M}_{\phi}^2 - \mathcal{M}_s^2} \quad . \quad (2.12)$$

In order to compute all the possible triple Higgs couplings (THCs), it is necessary to expand again the potential, but now in terms of the physical fields h and H , defined in Eq. 2.9. The coupling λ_{ijk} is defined such that the Feynman rules are given by [20]:

$$V_{ijk} = \begin{array}{c} \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{array} = -i v n! \lambda_{ijk} \quad , \quad (2.13)$$

where n is the number of identical particle in the vertex, and $i, j, k \in \{h, H\}$. Then, the THC equations for all combinations of the Higgs bosons couplings are the following ones:

$$\begin{aligned}
\lambda_{hhh} &= \frac{\lambda_{Hv}}{2} c_\alpha^3 + \left(\frac{\kappa_{SH}}{2} + \frac{\lambda_{SHvS}}{2} \right) c_\alpha^2 s_\alpha + \frac{\lambda_{SHv}}{2} s_\alpha^2 c_\alpha + \left(\frac{\kappa_S}{3} + 2\lambda_{SvS} \right) s_\alpha^3, \\
\lambda_{hhH} &= \left(\frac{\kappa_{SH}}{2} + \frac{\lambda_{SHvS}}{2} \right) c_\alpha^3 + \left(\lambda_{SHv} - \frac{3}{2} \lambda_{Hv} \right) c_\alpha^2 s_\alpha \\
&\quad + (\kappa_S + 6\lambda_{SvS} - \kappa_{SH} - \lambda_{SHvS}) s_\alpha^2 c_\alpha - \frac{\lambda_{SHv}}{2} s_\alpha^3, \\
\lambda_{hHH} &= \frac{\lambda_{SHv}}{2} c_\alpha^3 + (\kappa_S - \lambda_{SHvS}) c_\alpha^2 s_\alpha + \left(\frac{3\lambda_{Hv}}{2} - \lambda_{SHv} \right) s_\alpha^2 c_\alpha + \frac{\lambda_{SHvS}}{2} s_\alpha^3, \\
\lambda_{HHH} &= \frac{\kappa_S}{3} c_\alpha^3 - \frac{\lambda_{SHv}}{2} c_\alpha^2 s_\alpha + \frac{\lambda_{SHvS}}{2} s_\alpha^2 c_\alpha - \frac{\lambda_{Hv}}{2} s_\alpha^3.
\end{aligned} \tag{2.14}$$

The couplings of the Higgs bosons to any SM particle are modified with respect to the corresponding SM Higgs boson coupling. In the RxSM, the couplings are suppressed compared to the Higgs boson's couplings with SM particles by the cosine of the mixing angle for the light Higgs

boson and by the sine of the mixing angle for the heavy Higgs boson as,

$$g_{hi}^{\text{RxSM}} = g_{\phi i}^{\text{SM}} \cdot \cos \alpha \quad g_{Hi}^{\text{RxSM}} = -g_{\phi i}^{\text{SM}} \cdot \sin \alpha \quad (2.15)$$

Correspondingly, the decay rates of the RxSM Higgs bosons h and H , respectively, to the SM particles change as

$$\Gamma_{h \rightarrow ii}^{\text{RxSM}} = \Gamma_{h \rightarrow ii}^{\text{SM}} \cdot \cos^2 \alpha \quad \Gamma_{H \rightarrow ii}^{\text{RxSM}} = \Gamma_{H \rightarrow ii}^{\text{SM}} \cdot \sin^2 \alpha \quad (2.16)$$

where $\Gamma_{h,H \rightarrow ii}^{\text{SM}}$ denotes the decay width of the corresponding Higgs boson in SM framework. Although the decay width of the heavy Higgs boson to SM particles has been considered, it must be taken into account the possible decay of this heavy Higgs boson into two the light Higgs bosons. The total decay width of the heavy Higgs boson can additionally be modified by the possible decay into two light Higgs-bosons if kinematically allow, which (for $m_H > 2m_h$) is given by [21]

$$\Gamma_{H \rightarrow hh} = \frac{\lambda_{hhH}^2}{32\pi m_H} \sqrt{1 - \frac{4m_h^2}{m_H^2}} \quad (2.17)$$

Therefore, the total decay width of the heavy Higgs boson is computed as

$$\Gamma_{H,\text{tot}} = \Gamma_{H \rightarrow ii}^{\text{RxSM}} + \Gamma_{H \rightarrow hh} = \Gamma_{H \rightarrow ii}^{\text{SM}} \cdot \sin^2 \alpha + \frac{\lambda_{hhH}^2}{32\pi m_H} \sqrt{1 - \frac{4m_h^2}{m_H^2}} \quad (2.18)$$

For $\cos \alpha \rightarrow 1$, the light Higgs boson h has the same properties as predicted by the SM with $\Gamma_h^{\text{RxSM}} = \Gamma_H^{\text{SM}}$ (for $m_h = m_H^{\text{SM}} = 125.1$ GeV), which is known as the alignment limit. In case of $\cos \alpha \rightarrow 0$, H has SM-like couplings and $\Gamma_H^{\text{RxSM}} = \Gamma_H^{\text{SM}}$ for $\Gamma_{H \rightarrow hh} = 0$ (and $m_H = m_H^{\text{SM}} = 125.1$).

Furthermore, one can eliminate the dimensionless Lagrangian parameters λ_H , λ_S and λ_{SH} in favor of the two Higgs bosons masses $m_{h,H}^2$ and the mixing angle α as in [19]:

$$\lambda_H = \frac{1}{2v^2} [m_h^2 + m_H^2 + (m_h^2 - m_H^2) \cos(2\alpha)] \quad , \quad (2.19)$$

$$\lambda_S = \frac{1}{8v_S^3} [\kappa_{SH} v^2 + v_S(m_h^2 + m_H^2 - 2v_S \kappa_S) - (m_h^2 - m_H^2) v_S \cos(2\alpha)] \quad , \quad (2.20)$$

$$\lambda_{SH} = \frac{1}{2vv_S} [(m_h^2 - m_H^2) \sin(2\alpha) - 2v \kappa_{SH}] \quad , \quad (2.21)$$

$$\tan 2\alpha = \frac{2\mathcal{M}_{\phi s}^2}{\mathcal{M}_{\phi}^2 - \mathcal{M}_s^2} = \frac{4vv_S(\kappa_{SH} - v_S \lambda_{SH})}{v^2 \kappa_{SH} - 2v_S(4v_S^2 \lambda_S + v_S \kappa_S - v^2 \lambda_H)} \quad . \quad (2.22)$$

In consequence, the scalar sector of the model can be expressed in terms of the following physical parameters:

$$m_h, m_H, \kappa_S, \kappa_{SH}, \alpha, v_S, M_W, M_Z, \alpha_{em}$$

It is possible to reduce more the number of free parameters of the model, by matching the light Higgs boson to the discovered SM-like Higgs boson at the LHC and fixing the experimental vacuum expectation value v ² [22] and the mass of the light Higgs boson mass m_h [23], according to the experimental measurements made at the LHC. This leaves the model with five free parameters.

$$m_h = 125.11 \text{ GeV}, \quad v \sim 246 \text{ GeV}$$

2.2 Theoretical and Experimental Constraints

When studying the model, both theoretical and experimental constraints must be taken into account. This subsection briefly summarized the constraints considered throughout the analysis.

Theoretical Constraints

As it was mentioned in the introduction (Chap. 1), it must be ensured that the potential is bounded from bellow and the couplings are within the perturbative regime, following the references [16] and [17]. Firstly, for the scalar potential to be stable, the quartic couplings must be positive in all directions of the fields. This is translated into a requirement of a positive Hessian determinant of the potential with respect to the fields ϕ and s :

$$\begin{vmatrix} \partial_{\phi\phi} V & \partial_{s\phi} V \\ \partial_{\phi s} V & \partial_{ss} V \end{vmatrix} > 0 \quad . \quad (2.23)$$

Secondly, to ensure the perturbative couplings, we require $\frac{\lambda_H}{2}$, $\frac{\lambda_{SH}}{2}$ and $\frac{\lambda_S}{2}$ to be smaller than 4π . Taking all these theoretical conditions into account, the following theoretical constraints are found [16]:

$$0 < \lambda_H, \lambda_{SH} \quad \& \quad \frac{\lambda_H}{2}, \frac{\lambda_{SH}}{2}, \frac{\lambda_S}{2} < 4\pi \quad \& \quad \lambda_{SH} > -2\sqrt{\lambda_H \lambda_S} \quad . \quad (2.24)$$

As it will be seen later, phenomenological restrictions even yield $\lambda_S < 0.5$. As pointed out in [16], unitarity constraints do not yield further restrictions on the parameter space beyond Eq. 2.24.

²Note that the Vacuum Expectation Value is calculated from Fermi's constant G_F as

$$v = \left(\sqrt{2} G_F \right)^{-1/2}$$

Constraints from direct Higgs-boson searches at colliders

The exclusion limits at the 95% C.L. of all relevant BSM Higgs boson searches (including Run 2 data from the LHC) are included in the public code `HiggsBounds v.6` ([24], [25], [26], [27], [28]), which is included in the public code `HiggsTools` [29]. For a parameter point in a particular model, `HiggsBounds` determines on the basis of expected limits which is the most sensitive channel to test each BSM Higgs boson. Then, based on this most sensitive channel, `HiggsBounds` determines whether the point is allowed or not at the 95% CL. As input, `HiggsBounds` requires some specific predictions from the model, like branching ratios or Higgs-boson couplings.

Constraints from the properties of the ~ 125 GeV Higgs boson

Any model beyond the SM must accommodate a Higgs boson with mass and signal strengths as they were measured at the LHC. For the parameter points used, the compatibility of the $\mathcal{CP}$ -even scalar h with a mass of 125.09 GeV, h_{125} , with the measurements of signal strengths at the LHC is tested with the code `HiggsSignals v.3` ([30], [26], [31] and [29]), which is included in the code `HiggsTools`. The code provides a χ^2 value for the h_{125} predictions of a given model in comparison to the measurements of the Higgs-boson signal rates and masses from the LHC. To determine the allowed parameter space of the RxSM, we require that

$$\chi_{h_{125}, \text{RxSM}}^2 - \chi_{h_{125}, \text{SM}}^2 < 6.3 \quad (\text{with } \chi_{h_{125}, \text{SM}}^2 = 159.7) ,$$

where this upper limit corresponds to the allowed confidence region at the 95% CL for two degrees of freedom.

2.3 Alignment limit

In order to diagonalize the Higgs boson mass matrix (Eq. 2.8), two alternative ways that define two concrete limits of the model can be used: the alignment without decoupling and alignment with decoupling.

2.3.1 Alignment without decoupling

In this limit, the diagonalization is done by imposing the non-diagonal terms of $\mathcal{M}^2$ from Eq. 2.8 equal to zero:

$$\mathcal{M}_{\phi s}^2 = v(\kappa_{SH} + \lambda_{SH} v_S) = 0 \tag{2.25}$$

If one introduces this imposition into the Eq. 2.12, this leads to a mixing angle $\alpha = 0$. For this value of the mixing angle, the couplings between SM particles and the heavy Higgs boson disappears, nearly recovering the SM theory of the Higgs sector. Regarding to the THCs, it is clearly evident

that if the mixing angle is equal to zero, λ_{hhh} tends to the SM value, but for λ_{hhH} , it can be seen that if the mixing angle is set to zero, the coupling remains as:

$$\lambda_{hhH} = \frac{\kappa_{SH} + \lambda_{SH}v_S}{2} \quad (2.26)$$

Taking into account the previous consideration of the non-diagonal term and knowing that v is a non zero term, one can clearly see that this coupling goes exactly to zero, recovering the SM. In this scenario, the BSM contribution to the double light Higgs production ($pp \rightarrow H \rightarrow hh$) cross section at tree level becomes zero. However, the THCs involving two or three heavy Higgs bosons (λ_{hHH} and λ_{HHH}) do not vanish.

2.3.2 Alignment with decoupling

Another option for the diagonalization of the mass matrix is making the singlet mass term m_s^2 much larger than the off diagonal terms. This second limit is known as the alignment with decoupling limit:

$$\frac{\mathcal{M}_{\phi s}^2}{\mathcal{M}_s^2} \ll 1 \quad (2.27)$$

If one applies this limit into the Eq. 2.11, is straightforward that a decoupling of the masses occur, giving the $\mathcal{M}_\phi^2 \equiv m_h^2$ and $\mathcal{M}_s^2 \equiv m_H^2$. It is also clear to verify that in this limit the mixing angle goes to zero, recovering the SM.

$$\tan 2\alpha = \frac{2\mathcal{M}_{\phi s}^2}{\mathcal{M}_\phi^2 - \mathcal{M}_s^2} \approx \frac{2\mathcal{M}_{\phi s}^2}{\mathcal{M}_s^2} \ll 1 \quad (2.28)$$

As it was seen in the alignment without decoupling, λ_{hhh} approaches the SM value. In contrast, the λ_{hhH} does not vanish, since the non-diagonal term of the mass matrix is not zero and there is no $\mathcal{M}_s^2$ dividing in the expression (See Eq. 2.26)

In this limit, there are two possible ways for getting the alignment without decoupling. The first option is to decrease $\mathcal{M}_{\phi s}^2$ term in the numerator, which will have similar results to the alignment without decoupling. However, the second option, which consists in increasing the value of $\mathcal{M}_s^2$, is much more interesting due to what implies. If observing the expression of the mass of the singlet, one can see that depends on many Lagrangian parameters:

$$\mathcal{M}_s^2 = v_S(4\lambda_S v_S + \kappa_S) - \frac{\kappa_S v^2}{2v_S} \quad (2.29)$$

Since λ_S is bounded due to the unitarity, it can not increased too much for increasing $\mathcal{M}_s^2$. Then, the parameter κ_S is not prevented from increasing, so the limit of the alignment without decoupling

can be achieved by tending this value to infinity. Increasing κ_S is interesting due to the fact that this Lagrangian parameter comes from the S^3 term, which comes from not imposing the $\mathbb{Z}_2$ symmetry.

In order to understand if the model really decouples, one has to observe the contribution of the heavy Higgs boson to the double Higgs production, which must tend to zero. When computing the resonant s -channel diagram (Diagram (b) from Fig. 3.1), it can be shown that the propagator of the studied diagram is proportional to λ_{hhH} divided by m_H^2 , thus, this division in the alignment limit holds as:

$$\frac{\lambda_{hhH}}{m_H^2} \sim \frac{\kappa_{SH} + \lambda_S v_S}{2m_H^2} = \frac{\mathcal{M}_{\phi s}^2}{2v m_H^2} \sim \frac{\mathcal{M}_{\phi s}^2}{2v \mathcal{M}_s^2} \ll 1 \ .$$

Consequently, if the condition of Eq. 2.27 holds, the contribution of the BSM triangular diagram to the di-Higgs production cross section tends to zero.

2.3.3 Benchmark Plane

After using the tadpoles equations for the minimization conditions, the model has seven free parameters, but after fixing the mass of the light Higgs boson and the vacuum expectation value at the end of the Sec. 2.1, the number of free parameters is reduced to five:

$$\alpha \ , \ v_S \ , \ m_H \ , \ \kappa_S \ , \ \kappa_{SH}$$

In [16], a complete scan parameter of the RxSM model was carried in order to address what points achieved the condition to have a SFOEWPT. Out of the points that have a SFOEWPT, they kept eight points that maximizes $\sigma(pp \rightarrow H \rightarrow hh)$ from which a plane that drives a SFOEWPT is defined from three phenomenological conditions. This three phenomenological conditions are the next ones (as in [12]):

$$\begin{aligned} \kappa_{SH} v_S &= -16000 \ , \\ \lambda_H &= 0.36 \ , \\ \kappa_S &= -560 \sqrt{2\lambda_S} \ . \end{aligned}$$

With these set of conditions, a plane defined using only two free parameters (λ_S and v_S) can be created whose points allows a SFOEWPT. Throughout this analysis two projections will be used: the first one is defined by the two free parameters in the Lagrangian basis λ_S and v_S and the second plane is the projection to λ_{hhH} and $\kappa_\lambda = \lambda_{hhH}^{\text{RxSM}} / \lambda_{hhH}^{\text{SM}}$, where $\lambda_{hhH}^{\text{SM}}$ is computed by setting the mixing angle to zero (alignment limit).

In Fig. 2.1 and Fig. 2.2 the prediction of the mixing angle and the heavy Higgs boson mass is shown for the planes v_S - λ_S and λ_{hhH} - κ_λ . Regarding the mixing angle, this angle is clearly

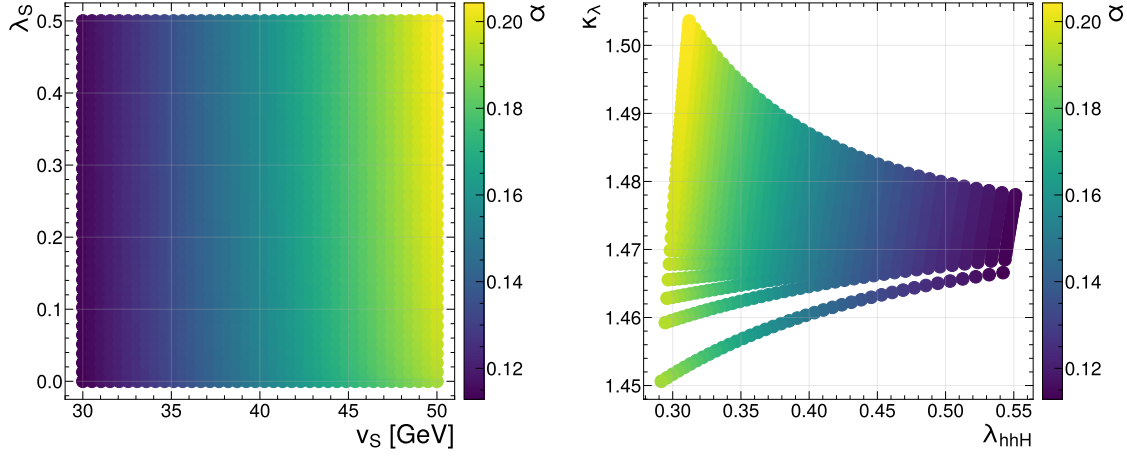


Figure 2.1: Mixing angle α variation in the benchmark plane. Left: Variation in the v_S - λ_S plane. Right: Variation in the λ_{hhH} - κ_λ plane.

proportional to the vacuum expectation values of the BSM heavy Higgs boson, while inversely proportional to the BSM THC λ_{hhH} , where the different values are within the range of $\sim [0.12, 0.20]$. On the other hand, the heavy Higgs mass is seen to be inversely proportional to the singlet VEV and proportional to the λ_{hhH} THC. All this in a range of $[450, 750]$ GeV. In Fig. 2.3, the prediction of the decay width of the heavy Higgs boson is studied and, as well as for the heavy Higgs boson mass, the decay width is inversely proportional to the singlet vacuum expectation value and directly proportional to the BSM THC λ_{hhH} .

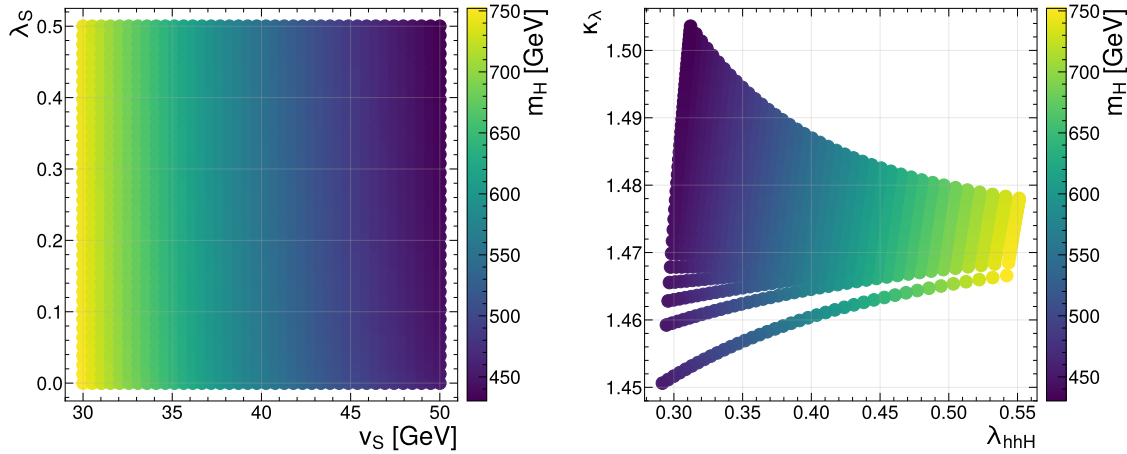


Figure 2.2: Heavy Higgs mass m_H variation in the benchmark plane. Left: Variation in the v_S - λ_S plane. Right: Variation in the λ_{hhH} - κ_λ plane.

In the following four figures (Figs. 2.4, 2.5, 2.6 and 2.7) the prediction of the available THCs within the RxSM are studied in both v_S - λ_S and λ_{hhH} - κ_λ . First of all, the strength of the trilinear SM-like Higg coupling is presented, which is show to increase as the singlet vacuum expectation value and λ_S increases. For the cases where the BSM Higgs boson enters into the trilinear coupling, this effect completely vanishes. While for the λ_{hhH} and λ_{hHH} couplings the same tendency is shown,

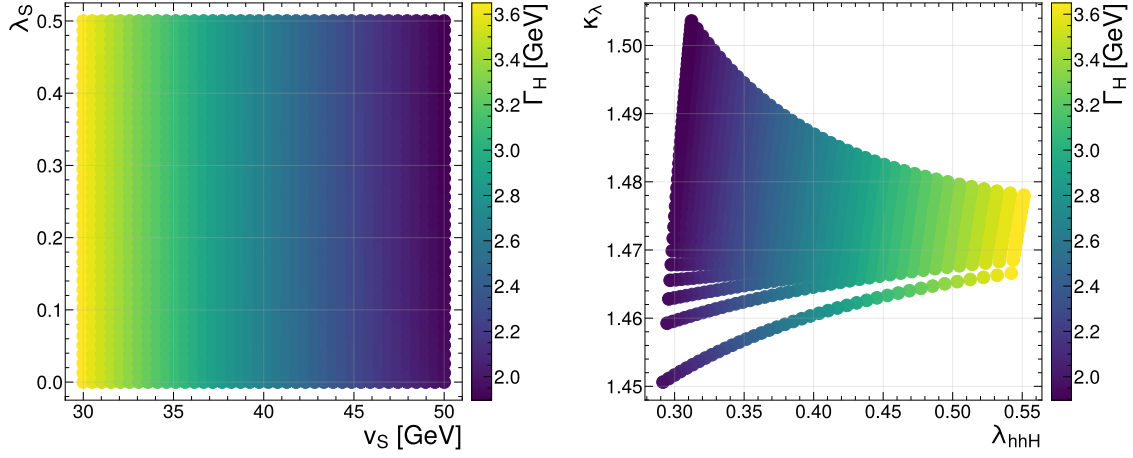


Figure 2.3: Heavy Higgs decay width Γ_H variation in the benchmark plane. Left: Variation in the v_S - λ_S plane. Right: Variation in the λ_{hhH} - κ_λ plane.

which is inversely proportional to the singlet VEV, λ_{hhH} also seems to demonstrate a relation with increasing the λ_{hhH} coupling. However, the coupling composed entirely by heavy Higgs bosons shows completely the contrary tendency as κ_λ .

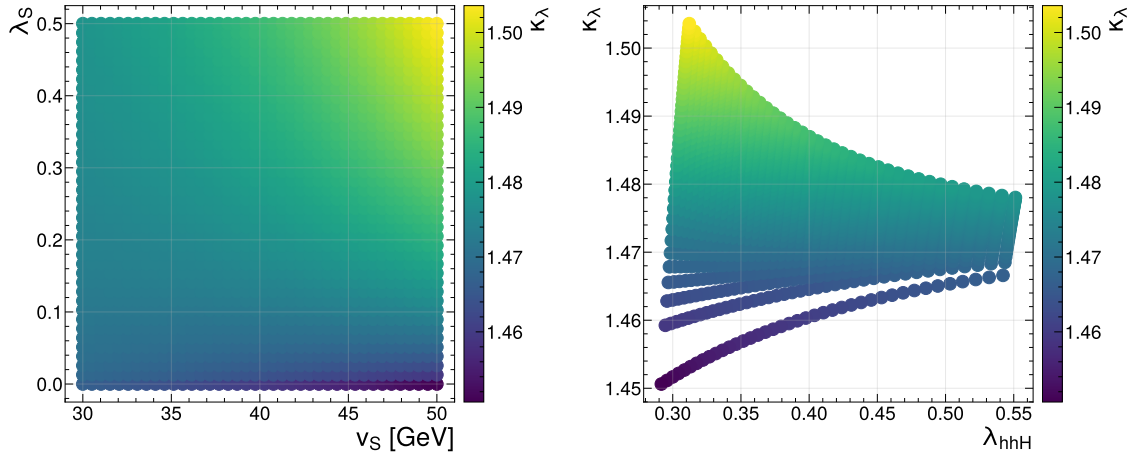


Figure 2.4: SM-like THC λ_{hhh} variation in the benchmark plane. Left: Variation in the v_S - λ_S plane. Right: Variation in the λ_{hhH} - κ_λ plane.

Finally, in Fig. 2.8, the prediction of the total cross section for the production of two light Higgs bosons at Leading Order (LO) in both v_S - λ_S and λ_{hhH} - κ_λ planes is shown. Details will be given in the next section. This observable tends to increase for values where λ_S and v_S are high, but also for values where the strength of the trilinear SM-like Higgs coupling increases in the plane.

Among all the points inside the benchmark plane, through the analysis that will be carried out, only eight of them will be analyzed. These points coincide with the points defined in [12] and their parameters are summarized in Tab. 2.1. The necessary input parameters and other relevant quantities are given in Tab. 2.1 and their location in the λ_{hhH} - κ_λ plane are presented in Fig. 2.9.

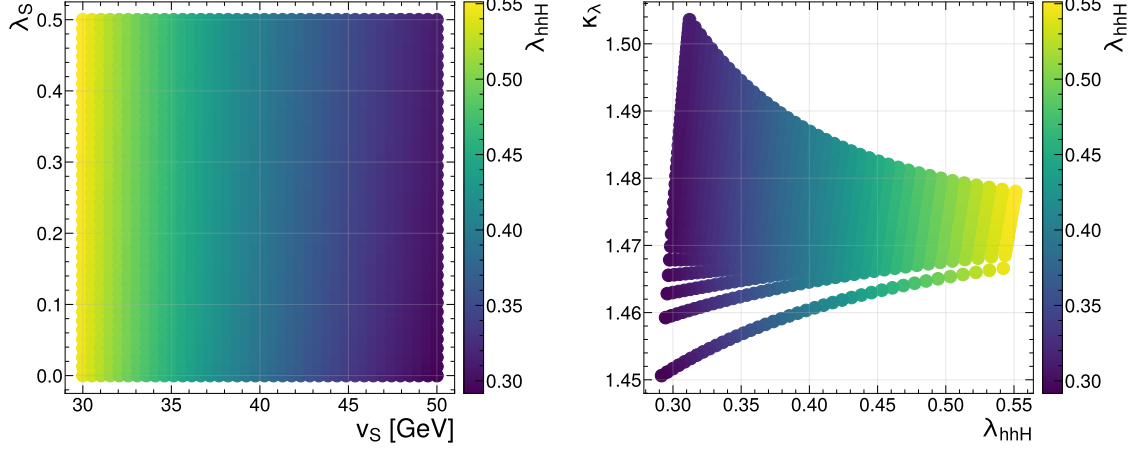


Figure 2.5: BSM THC λ_{hhH} variation in the benchmark plane. Left: Variation in the v_S - λ_S plane. Right: Variation in the λ_{hhH} - κ_λ plane.

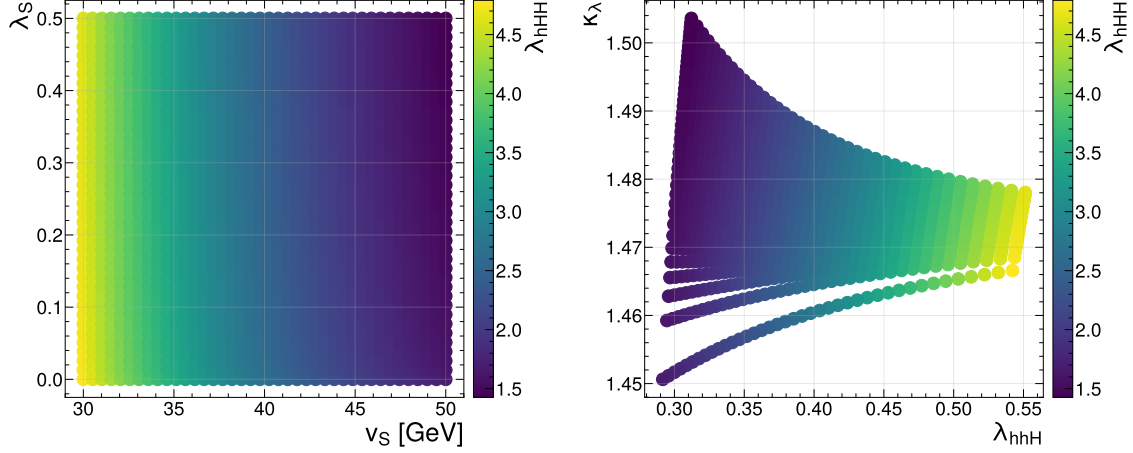


Figure 2.6: BSM THC λ_{hHH} variation in the benchmark plane. Left: Variation in the v_S - λ_S plane. Right: Variation in the λ_{hhH} - κ_λ plane.

BP	m_H [GeV]	α	v_S [GeV]	λ_S	Γ_H [GeV]	κ_λ	λ_{hhH}	λ_{hHH}	λ_{HHH}
1	459.2	0.984	46.3	0.445	1.967	1.473	0.355	1.680	-3.934
2	464.9	0.985	46.3	0.225	1.977	1.463	0.349	1.734	-3.549
3	469.4	0.985	47.4	0.000	1.925	1.435	0.326	1.813	-2.279
4	529.8	0.988	41.9	0.000	2.294	1.439	0.380	2.340	-2.898
5	577.5	0.990	37.5	0.390	2.624	1.459	0.448	2.781	-5.202
6	531.7	0.988	40.8	0.225	2.366	1.457	0.404	2.332	-4.250
7	642.9	0.992	34.2	0.055	2.926	1.450	0.486	3.482	-5.096
8	657.9	0.993	33.1	0.390	3.039	1.458	0.511	3.636	-6.408

Table 2.1: Benchmark points which produces a SFOEWPT and are going to be studied in this thesis as in [32].

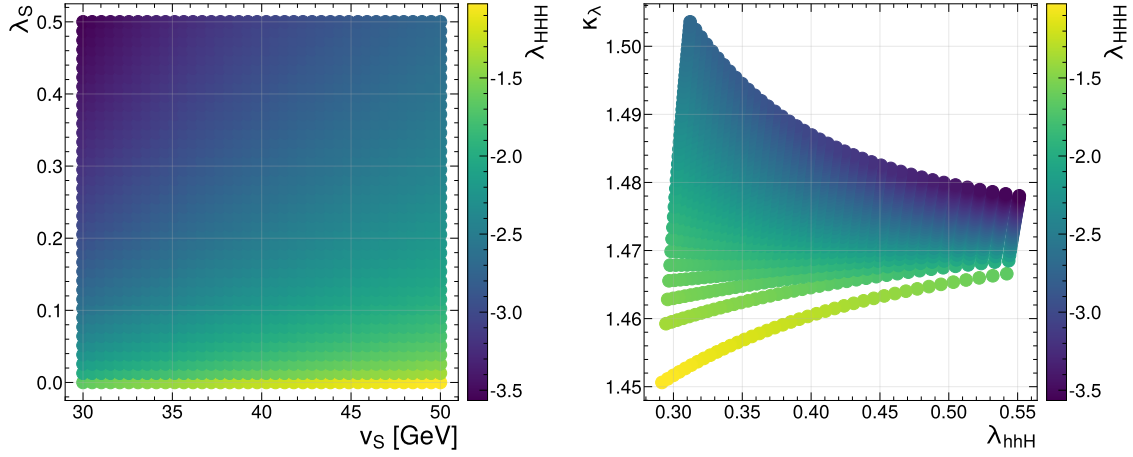


Figure 2.7: BSM THC λ_{HHH} variation in the benchmark plane. Left: Variation in the v_S - λ_S plane. Right: Variation in the λ_{hhH} - κ_λ plane.

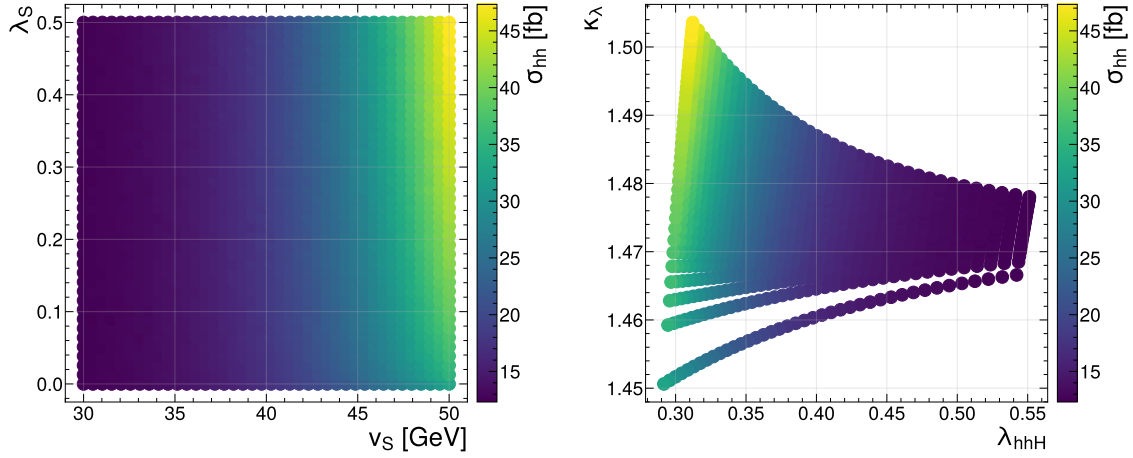


Figure 2.8: Total cross section at LO σ_{hh}^{LO} variation in the benchmark plane. Left: Variation in the v_S - λ_S plane. Right: Variation in the λ_{hhH} - κ_λ plane.

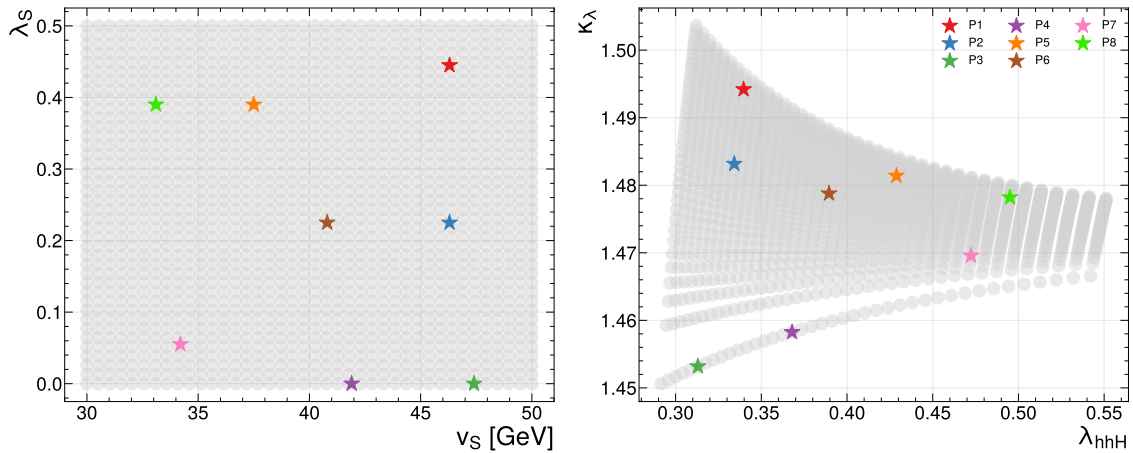


Figure 2.9: Position of the benchmark points from Tab. 2.1 in the v_S - λ_S and λ_{hhH} - κ_λ planes.

The analysis of the all di-Higgs production channels within the RxSM model has not been studied yet and it is an interesting research to test this new physics in the future high luminosity phase of the LHC, where the collected data will increase significantly.

In the LHC scenario, where proton-proton collisions occur, the production of double SM-like Higgs can not occur at tree-level and one has to go to one-loop diagrams. In the SM two diagrams contribute to the process: The box diagram (Diagram (c) in Fig. 3.1), where a heavy quark loop and two Yukawa couplings take part in, and the triangle diagram (Diagram (a) in Fig. 3.1), which has a heavy quark loop, a Yukawa coupling and a trilinear Higgs coupling. On the other hand, in the RxSM, the same two diagrams contribute to the di-Higgs production, but they can differ from the SM by the Yukawa coupling and the trilinear Higgs couplin λ_{hhh} . In addition, another triangle diagram plays an important role in the cross section contribution, where a BSM Higgs (heavy Higgs) boson propagates (Diagram (b) in Fig. 3.1). In this new diagram, the Yukawa coupling and the trilinear Higgs coupling are BSM couplings specific to the RxSM.

Although the RxSM extends the diagrams that contributes to the di-Higgs production cross section, it also predicts new possible di-Higgs channels: hh , hH , HH . These new channels are referred to double Higgs productions where BSM Higgs bosons enters. In order to create these pairs of particles, the diagrams contributing to the double light Higgs boson production at one-loop level are considered, shown in Fig. 3.1. Nevertheless, the Yukawa couplings are changed compared to the previous diagrams, which will modulate the values of the cross sections.

In this thesis, both light and BSM di-Higgs production that RxSM model predicts are studied in order to explore completely the extended Higgs potential. The entire analysis of the properties of each of the diagrams have been studied using the open source code of **anyBSM** adopted to the studied RxSM model.

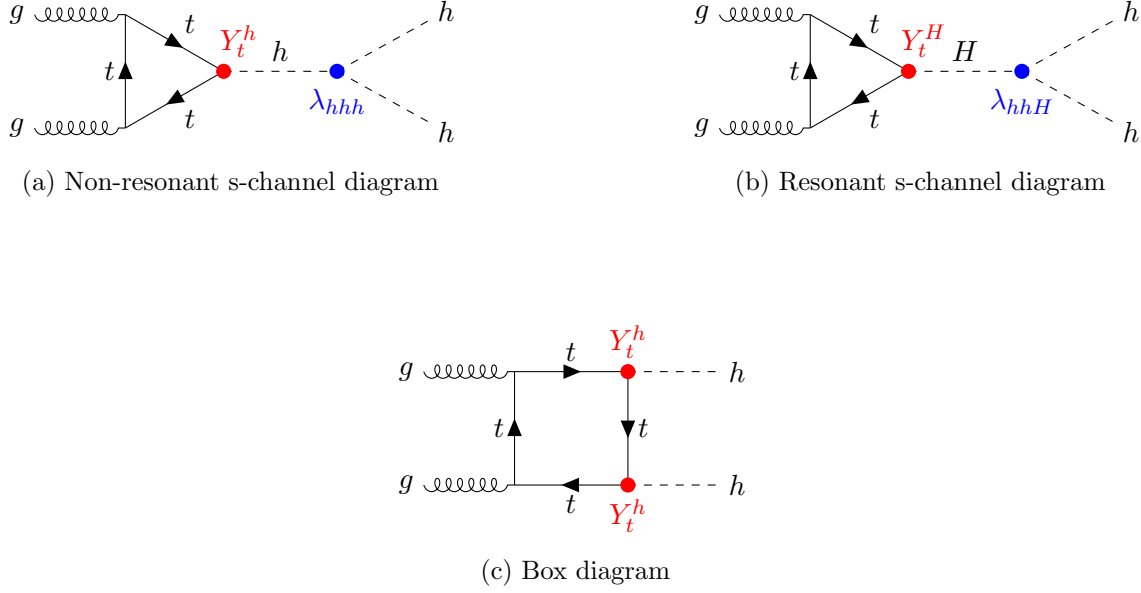


Figure 3.1: Feynman diagrams for the double SM-like Higgs boson production ($gg \rightarrow hh$) in the RxSM model via gluon fusion. (a) and (b) are the contributions associated to a triangular heavy quarks loop contribution with an exchange of a Higgs boson (h or H) in the s -channel and (c) is the contribution to the heavy quark box loop.

3.1 *anyBSM: anyHH and anyH3*

anyBSM is a public `Python` code composed of several modules designed to calculate higher-order corrections in both SM and BSM physics [19]. This code provides with precision calculations, such as for total cross section for di-Higgs production processes and trilinear Higgs couplings. To support some BSM models, for example the Minimal Supersymmetric Standard Model (MSSM) or the RxSM, **anyBSM** relies on the Universal FeynRules Output (UFO) format. By importing model definitions in this standardized Python-encoded format, the tool automatically processes the necessary fields, parameters, and Feynman rules. The code architecture is interconnected with object-oriented classes. Furthermore, it is linked to the `Fortran` based library `COLLIER` to perform the numerical evaluation of the loop integrals.

Among the available scripts within the package, two specific modules were utilized in this thesis: **anyHH** and **anyH3**. Within the **anyHH** module, both total and differential cross sections for di-Higgs production were computed for different scenarios, having the possibility of isolating specific diagrams contributions. For all evaluations performed with **anyHH**, results were obtained at leading order (LO) in QCD. Conversely, the **anyH3** module was employed to extract the tree-level values of the THCs, although the framework is inherently capable of computing these couplings at the full one-loop level.

3.2 Parton Distribution Functions

In proton-proton collisions at the LHC at high center-of-mass energies, the fundamental interactions do not occur between hadrons as whole entities, but rather between the partons (quarks and gluons)

that constitute them. At these extreme energies, parton interactions are not limited to the valence quarks; instead, they involve a complex, dynamic sea of virtual quarks and gluons.

This internal structure is exactly what makes cross-section calculations much more complicated for proton colliders. Unlike collisions between fundamental particles where the interaction energy is precisely known, the exact energy carried by the individual partons that actually collide inside the proton cannot be determined beforehand. To solve this, we have to integrate over all possible energies. Therefore, Parton Distribution Functions (PDFs) are used to describe the probability density of finding a specific parton flavour i carrying a specific longitudinal momentum fraction x of the parent hadron, evaluated at a factorization scale μ_F^2 .

Since the dominant production mechanism for the di-Higgs processes investigated in this work is gluon-gluon fusion ($gg \rightarrow hh, hH, HH$), the gluon PDF plays a critical role, as the hadronic cross section relies heavily on the gluon-gluon parton luminosity. Out of the various PDF sets available in the literature, the PDF4LHC15_nlo_mc [33] set has been adopted throughout this analysis. This ensures a standardized treatment of PDF uncertainties and a strong coupling value of $\alpha_s = 0.119$, which is compatible with next-to-leading order (NLO) QCD predictions.

3.3 Branching Ratios

Many of the particles in the SM are unstable and can not be detected directly. Instead, they must be measured indirectly through their decay products. The fundamental parameter controlling this behavior is the total decay width, Γ_{tot} , which is inversely proportional to the particle's mean lifetime. When a particle can decay into multiple independent channels, a partial decay width, Γ_i , is assigned to each specific decay mode i , such that the total decay width is the sum of all possible channels. However, when studying a specific particle, it is not possible to observe all the decay modes and the most probable ones are studied. To quantify the probability of a particle decaying into a certain final state, the Branching Ratio (BR) is defined as the ratio of the partial decay width to the total decay width:

$$\text{BR}(X \rightarrow YZ) = \frac{\Gamma(X \rightarrow YZ)}{\Gamma_{\text{tot}}} , \quad (3.1)$$

where X is the initial state and Y and Z are the products of the decay.

The processes that are studied throughout this thesis provide two scalar particles, each of them being a light Higgs boson or a heavy Higgs boson. However, these type of scalar particles are not detectable in the LHC, thus their products must be studied.

Many researches have been performed to understand and give values to the branching ratios of the Higgs boson. In [34], an analysis of the Higgs decays was performed where a value to each of the BR was set for different values of the Higgs mass. As it can be seen in Fig. 3.2, the importance of each of the modes fluctuates as the Higgs boson mass changes.

In Tabs. 3.1 and 3.2, the main decay channel with their respective branching ratios are shown. The first table is referred to the SM Higgs boson branching ratios, while the second table mentions

m_h [GeV]	$h \rightarrow b\bar{b}$	$h \rightarrow WW$	$h \rightarrow ZZ$	$h \rightarrow \gamma\gamma$
125.1	0.575	0.216	0.0267	0.0023

Table 3.1: Branching ratios of the most relevant decays for the SM Higgs at $m_h = 125.1\text{GeV}$ [34].

the decays width for heavier Higgs boson which coincide with the masses of the masses of the heavy Higgs bosons of each of the benchmark points studied in this work.

m_h [GeV]	$h \rightarrow WW$	$h \rightarrow ZZ$	$h \rightarrow t\bar{t}$
459.2	0.549	0.259	0.191
464.9	0.549	0.259	0.191
469.4	0.549	0.259	0.191
529.8	0.547	0.263	0.190
577.5	0.555	0.270	0.175
531.7	0.549	0.265	0.185
642.9	0.566	0.277	0.156
657.9	0.570	0.280	0.150

Table 3.2: Branching ratios of the most relevant decay for the heavy Higgs of the benchmark points for different Higgs masses. Data obtained from [34].

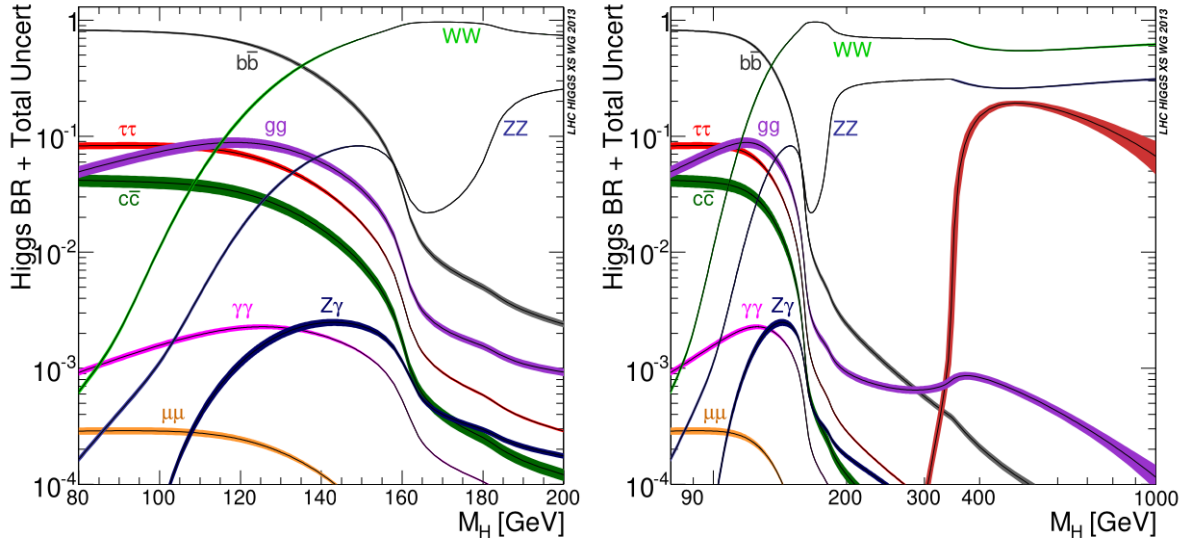


Figure 3.2: Higgs branching ratios and their uncertainties for the low mass range (left) and for the full mass range (right). Figure obtained from [34].

To calculate the expected number of events for these processes, the BR of the scalar bosons must be determined. In principle, the sum of all possible branching ratios for a given unstable particle ϕ (where $\phi = h, H$) to any final state X is strictly constrained by probability conservation to be

equal to unity. However, in this study, only the most relevant decay modes are taken into account. Therefore, the total branching ratio for each of the studied scalar bosons will be:

$$\text{BR}(h) = \text{BR}(h \rightarrow b\bar{b}) + \text{BR}(h \rightarrow WW) + \text{BR}(h \rightarrow gg) + \text{BR}(h \rightarrow \gamma\gamma) , \quad (3.2)$$

$$\text{BR}(H) = \text{BR}(H \rightarrow WW) + \text{BR}(H \rightarrow ZZ) + \text{BR}(H \rightarrow t\bar{t}) , \quad (3.3)$$

where the values of each branching ratio is given in Tabs. 3.1 and 3.2.

3.4 Total cross section and Number of expected events

After utilizing **anyBSM** to compute the total di-Higgs production cross section at NLO accuracy for the chosen PDF set, the expected deviation from the SM prediction can be evaluated. This framework allows us to assess whether the BSM signal can be distinguished from the SM background at the HL-LHC, taking into account projected experimental uncertainties and assuming that theoretical predictions will achieve the required precision by the start of the HL-LHC phase.

For estimating the number of expected events that it is hoped to be observed once the HL-LHC phase ends, the integrated luminosity of this phase has to be considered. For this case, an integrated luminosity of $\mathcal{L}_{\text{int}} = 3 \text{ ab}^{-1}$ is expected, as mentioned in [35]. Since this work is based on the analysis of the di-Higgs production in the RxSM, the number of expected events for this process is calculated following the next equation:

$$\hat{N}_{pp \rightarrow \phi\phi} = \sigma_{pp \rightarrow \phi\phi} \times \mathcal{L}_{\text{int}} \times \text{BR}(\phi_i) \times \text{BR}(\phi_j) , \quad (3.4)$$

where $\phi_i, \phi_j \in \{h, H\}$, $\mathcal{L}_{\text{int}}$ in the mentioned integrated luminosity and $\sigma_{pp \rightarrow \phi\phi}$ is the total cross section of the desired processes, where the available options are $\{hh, hH, HH\}$. For calculating the uncertainty of $\hat{N}_{pp \rightarrow \phi\phi}$ in the HL-LHC, the anticipated significance of the SM di-Higgs production cross section at this phase of the experiment will be taken which is $s_{\text{SM}} = 7.6\sigma$ [36] and it will be rescaled taking into account that the number of expected events are different. Since a Gaussian distribution is assumed, the error scales as the square root of the number of events, then error in the number of expected events for a given processes is rescaled as

$$\Delta \hat{N}_{pp \rightarrow \phi_i \phi_j} = \frac{\hat{N}_{pp \rightarrow hh}^{\text{SM}}}{s_{\text{SM}}} \times \sqrt{\frac{\hat{N}_{pp \rightarrow hh}^{\text{SM}}}{\hat{N}_{pp \rightarrow \phi_i \phi_j}}} , \quad (3.5)$$

where $\hat{N}_{pp \rightarrow hh}^{\text{SM}}$ are the number of expected events for the di-Higgs production within the SM. Using these last defined variables, one can estimate the expected significance of the signal of the different di-Higgs productions at the HL-LHC. This measure is quantified by the ratio between the number of expected events to its uncertainty, which provides a measure of the signal's strength:

$$Z = \frac{\hat{N}_{pp \rightarrow \phi_i \phi_j}}{\Delta \hat{N}_{pp \rightarrow \phi_i \phi_j}} . \quad (3.6)$$

4

Results

In this section, the results of the phenomenological analysis are presented. First, we address the initial discrepancies encountered during the validation phase of our theoretical framework. Following this, the total cross sections for the different di-Higgs production channels within the RxSM are presented, alongside the expected signal yields for each decay mode. Finally, to constrain the THC's, a χ^2 fit is performed for each of the benchmark points defined in Tab. 2.1, assessing their projected limits for the High-Luminosity LHC phase.

4.1 Total Cross Section

As mentioned in Sec. 3.1, **anyBSM** provides the necessary framework to calculate all the observables of interest for this thesis. However, other public and private codes can evaluate these parameters for the same model. One such tool is **HPAIR**, a model-dependent code widely used in the literature [12, 37] to compute neutral Higgs boson pair production via gluon fusion.

To ensure reliability, an initial cross-validation of the cross sections at LO was performed between **anyBSM** and **HPAIR**. Initial comparisons for the RxSM revealed significant discrepancies. To rule out implementation errors, the Two-Higgs-Doublet Model (2HDM) was evaluated independently using both codes, yielding perfectly consistent results.

Returning to the RxSM, the differential cross sections were analyzed to try to infer the source of the disagreement.

In Fig. 4.1, we can see the differential distribution in the RxSM with different topologies of the invariant mass distribution. Specially, the distribution presents a dip around $m_{hh} \approx 290$ GeV, caused by the destructive interference between the light Higgs triangle diagram and the box diagram. In the Standard Model (SM), where $\kappa_\lambda = 1$, this interference occurs at roughly $m_{hh} \approx 250$ GeV. For the reference point evaluated here ($\kappa_\lambda \approx 1.5$), the dip shifts to $m_{hh} \approx 290$ GeV. Both codes successfully reproduced this specific interference threshold.

However, a second interference effect emerges at $m_{hh} \approx 460$ GeV, driven by the heavy Higgs

triangle diagram interacting with the box diagram. As shown in Fig. 4.1, the *red line* predicts a dip-peak structure. In contrast, the *blue line* predicts the inverse (a peak-dip structure). This specific topology is generated by the sign of the product between the heavy Higgs top-Yukawa coupling and the BSM trilinear coupling, $Y_t^H \times \lambda_{hhH}$ [38].

After revising the model, collaborators found an inconsistency in the implementation of top Yukawa coupling, Y_t^H . The authors of [38] demonstrated that flipping the sign of this product in the 2HDM significantly alters both the kinematic shape and the total integrated cross section. In Fig. 4.1, the differential cross section is computed by changing by hand the sign of the top quark Yukawa coupling. In addition, this sign flip induces a significant change in the value for the total cross section, changing from $\sigma = 76.12$ fb for a negative product ($Y_t^H \times \lambda_{hhH} < 0$) to $\sigma = 88.02$ fb for a positive product ($Y_t^H \times \lambda_{hhH} > 0$) for the first benchmark point from Tab. 2.1.

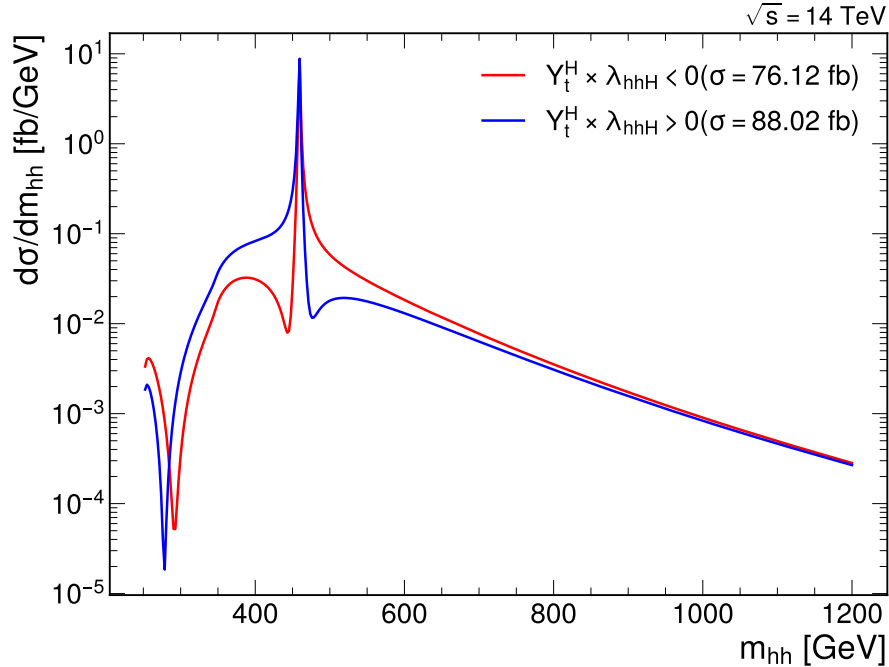


Figure 4.1: Differential cross section of the $pp \rightarrow hh$ process at $\sqrt{s} = 14$ TeV as a function of m_{hh} evaluated at BP1 for different signs of the $Y_t^H \times \lambda_{hhH}$ product. The *red* (*blue*) curve corresponds to a negative (positive) sign in the $Y_t^H \times \lambda_{hhH}$ product. The sign of the top Yukawa coupling is manually modified.

4.1.1 Parametric Impact on Di-Higgs Production

Once the reliability of the computational setup was established, the total NLO cross sections were evaluated for all possible channels (hh , hH , and HH). Fig. 4.2 illustrates the energy dependence of the cross section for BP1, indicating the center-of-mass energies of past LHC and the incoming HL-LHC runs and their associated cross sections within the investigated RxSM. In the SM, the

expected di-Higgs production cross section at 14 TeV is

$$\sigma_{\text{SM}} = 37.44 \text{ fb} ,$$

but the variation of κ_λ along the benchmark points and the introduction of the heavy Higgs triangle diagram modifies this observable. In the SM, if the κ_λ parameter is modified to larger values, the total cross section tends to decrease until reaching a minimum at $\kappa_\lambda \approx 2.4$ [39]. In contrast, the introduction of the heavy Higgs triangle diagram can increase or decrease the total cross section value, depending on the point of the benchmark plane that we are considering, as demonstrated in Tab. 4.1.

In the RxSM, as demonstrated in Sec. 2.3.3, values close to $\kappa_\lambda = 1.5$ tend to decrease the total cross section. This behavior arises because the interference between the light Higgs triangle diagram and the box diagram shifts the dip towards higher values of m_{hh} . At the same time, for large values of λ_{hhH} , which occur when the mass of the heavy Higgs boson is larger in the studied plane, the total cross section is suppressed compared to scenarios with smaller coupling values, due to the increase of the BSM Higgs boson mass. Therefore, within the selected benchmark plane, the heavy Higgs boson mass and the mixing angle¹ are the primary model parameters that dictate the variations in the total cross section for Higgs pair production.

BP	σ_{RxSM} [fb]	$\Delta\sigma/\sigma_{\text{SM}}$ [%]
1	76.12	103.31
2	70.25	87.63
3	63.18	68.75
4	42.32	13.03
5	34.81	-7.02
6	43.14	15.22
7	28.71	-23.31
8	27.90	-25.48

Table 4.1: Total cross section (σ_{RxSM}) and relative deviation from the SM prediction ($\Delta\sigma/\sigma_{\text{SM}}$) for the $pp \rightarrow hh$ channel at $\sqrt{s} = 14$ TeV for each benchmark point.

4.1.2 Expected Signal Yields

To contextualize the experimental viability of these signatures at the HL-LHC, $\hat{N}_{pp \rightarrow \phi_i \phi_j}$ (Eq. 3.4) was calculated for each of the studied productions and an integrated luminosity of $\mathcal{L}_{\text{int}} = 3 \text{ ab}^{-1}$. The most relevant decay modes of each Higgs boson were considered (branching ratios are detailed in Sec. 3.3). Signal yields and their respective statistical uncertainties were computed using Eq. 3.4 and Eq. 3.5.

The results, alongside the statistical significance (Z , defined in Eq. 3.6), are summarized in Tabs. 4.2, 4.3, and 4.4. The data reveals a remarkable contrast between the light di-Higgs channel (hh)

¹When you go closer to the alignment limit in the RxSM, the cross section gets closer to the SM value because the BSM physics is decoupled.

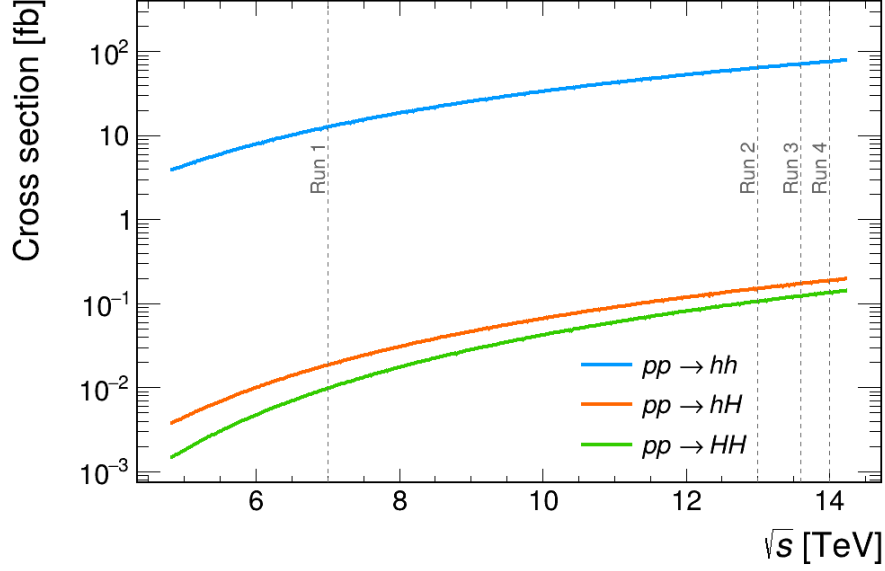


Figure 4.2: Total production cross section for the considered RxSM processes (hh , hH , HH) as a function of the center-of-mass energy $\sqrt{s}$ for BP1.

and the heavy final states (hH and HH).

The hh production cross sections are orders of magnitude larger than those of the heavy channels, whose cross sections do not surpass $\mathcal{O}(1\text{fb})$. The processes where two light Higgs bosons are produced becomes more relevant since the energy needed to produce is not that large as for the order processes, in addition, the heavy Higgs triangle diagram contributes significantly to the production to the hh production, which can enhance the total cross section for this specific process. This dominance is primarily driven by two factors. First, the kinematic threshold for hh production ($2m_h \approx 250$ GeV) is significantly lower than that of the hH and HH final states. Second, within the RxSM framework, the s -channel exchange of the heavy Higgs boson introduces a triangular diagram that contributes significantly to hh production, potentially providing a resonant enhancement to the total cross section for this specific process, depending on the selected benchmark point. Consequently, the processes where a heavy Higgs boson is produced in the final state yield very low significances. Due to the low significance values for these types of processes, from this point on, only the final states of two light Higgs bosons, hh , will be considered for future analysis.

4.2 Extraction of Triple Higgs Couplings

The total cross section measurements and statistical uncertainties of the di-Higgs production channels provide highly valuable information to access the values of the THCs. At the same time, differential cross sections become highly relevant in this analysis, as they reveal how variations in the triple Higgs couplings modify these distributions and, consequently, alter the total cross section value. In previous studies [12], the impact of the BSM THCs for the selected benchmark points was analyzed by studying both the total and differential cross sections. However, for obtaining their

BP	σ_{hh} [fb]	$\hat{N}_{pp \rightarrow hh}$	$\Delta \hat{N}_{pp \rightarrow hh}$	Z
P1	76.12	175485	7959	22.05
P2	70.25	161944	8285	19.55
P3	63.18	145641	8736	16.67
P4	42.32	97571	10674	9.14
P5	34.81	80247	11769	6.82
P6	43.14	99441	10573	9.41
P7	28.71	66174	12961	5.11
P8	27.90	64327	13145	4.89

Table 4.2: Cross section, expected number of events, statistical uncertainty, and significance for the hh production channel at the HL-LHC.

BP	σ_{hH} [fb]	$\hat{N}_{pp \rightarrow hH}$	$\Delta \hat{N}_{pp \rightarrow hH}$	Z
P1	0.188	495	149881	0.0033
P2	0.168	442	158669	0.0028
P3	0.147	387	169495	0.0023
P4	0.060	158	265074	0.0006
P5	0.032	85	361125	0.0002
P6	0.061	160	263580	0.0006
P7	0.014	37	547622	0.0001
P8	0.012	32	594108	0.0001

Table 4.3: Cross section, expected number of events, statistical uncertainty, and significance for the hH production channel at the HL-LHC.

BP	σ_{HH} [fb]	$\hat{N}_{pp \rightarrow HH}$	$\Delta \hat{N}_{pp \rightarrow HH}$	Z
P1	0.136	406	165400	0.0025
P2	0.119	357	176480	0.0020
P3	0.093	278	200123	0.0014
P4	0.055	166	258438	0.0006
P5	0.044	132	289928	0.0005
P6	0.064	191	241561	0.0008
P7	0.024	71	395525	0.0002
P8	0.023	68	405172	0.0002

Table 4.4: Cross section, expected number of events, statistical uncertainty, and significance for the HH production channel at the HL-LHC.

values and their respective statistical uncertainties, the following analysis strategy was followed.

As mentioned in the previous section, the significances of the BSM Higgs production processes (hH and HH) are too low to take those final states into account; therefore, the two light Higgs production and the THCs that are relevant at tree level will be studied. Those couplings are λ_{hhh} and λ_{hhH} , and their impact in modifying the invariant mass distributions will provide some clues about what is expected to be found in the HL-LHC.

To determine the upper and lower limits of each coupling, we performed a parameter scan over a 2D plane defined by κ_λ and λ_{hhH} , followed by a χ^2 minimization for the eight points considered in this study.

For doing this parameter scan, a random point generation have been done, taking into account the theoretical and experimental constraints (See Sec. 2.2). The points that were selected on each of the scenarios have been considered those that fulfil the imposed constraints. For each of the benchmark points, a 1% variations of m_H have been allowed corresponding to the expected accuracy with which a BSM scalar mass could be measured experimentally. The rest of the Lagrangian parameters entered in **anyBSM** (See Sec. 2.3.3) were varied within the following ranges:

$$v_S \in [1, 500] \text{ GeV} , \quad \kappa_S \in [-10000, 10000] \text{ GeV} , \quad \kappa_{SH} \in [-10000, 0] \text{ GeV} .$$

This iterative process yields a set of more than 35,000 points for each of the considered benchmark points within the entire 2D plane. However, for determine if the corresponding benchmark points would be a current expected measurable value in the HL-LHC, a restriction of 1% in the mixing angle was also imposed to constraint that free parameter.

For the extraction of the upper and lower bounds of each THC, 1σ (68%) and 2σ (95%) confidence levels have been used. Since only the $pp \rightarrow hh$ channel has been taken into account, two parameters are fitted. To complement the total cross-section input, it is assumed that κ_λ is determined separately, for instance, through a fit to the kinematic distributions. Consequently, in order to set contours, the χ^2 value is computed as follows:

$$\chi^2 = \left(\frac{\kappa_\lambda^{\text{BP}} - \kappa_\lambda}{\Delta\kappa_\lambda^{\text{BP}}} \right)^2 + \left(\frac{\sigma(pp \rightarrow hh)^{\text{BP}} - \sigma(pp \rightarrow hh)}{\Delta\sigma(pp \rightarrow hh)^{\text{BP}}} \right)^2 , \quad (4.1)$$

where $\kappa_\lambda^{\text{BP}}$ and $\sigma(pp \rightarrow hh)^{\text{BP}}$ are the values of the κ_λ and the total cross section of the studied benchmark point, respectively. In a two dimensional parameter space, the corresponding values for 1σ and 2σ are $\Delta\chi^2 = 2.30$ and $\Delta\chi^2 = 5.99$ [40].

By choosing $\sigma(pp \rightarrow hh)$ and κ_λ to extract the upper and lower limits of each of the THCs, we obtain information of the behavior of the λ_{hhh} coupling from the lower part of the m_{hh} distribution from the interference pattern of the triangle and box diagram, following the experimental analysis, while bounds for the λ_{hhH} can be extracted from the total cross section, $\sigma(pp \rightarrow hh)$.

For obtaining the errors of the considered parameters for the χ^2 fit, the statistical significance of the di-Higgs production in the SM of $s_{\text{SM}} = 7.6$ [36] is used and in order to calculate $\Delta\sigma(pp \rightarrow hh)^{\text{BP}}$,

BP	σ_{hh}^{BP} [fb]	$\Delta\sigma_{hh}^{\text{BP}}$ [fb]	$\kappa_\lambda^{\text{BP}}$	$\Delta\kappa_\lambda^{\text{BP}}$
1	76.12	3.48	1.473	0.555
2	70.25	3.63	1.463	0.551
3	63.18	3.83	1.435	0.540
4	42.32	4.68	1.439	0.542
5	34.81	5.16	1.459	0.549
6	43.14	4.64	1.457	0.549
7	28.71	5.69	1.450	0.546
8	27.90	5.76	1.458	0.549

Table 4.5: Values and uncertainties of the observables used for the χ^2 fit. The corresponding values and errors are computed from expected significances and uncertainties in the SM [36].

3.5 was followed

$$\Delta\sigma_{hh}^{\text{RxSM}} = \frac{\sigma_{hh}^{\text{SM}}}{s_{\text{SM}}} \sqrt{\frac{\sigma_{hh}^{\text{SM}}}{\sigma_{hh}^{\text{RxSM}}}} \quad , \quad (4.2)$$

and the κ_λ uncertainties from the left plot of Fig. 4 from [36] have been used to extrapolate the values of $\Delta\kappa_\lambda$ for each of the benchmark points. In Tab. 4.5, the uncertainties in κ_λ are shown for each of the considered benchmark points.

In Figs. 4.3, 4.4, 4.5 and 4.6, the distributions of the generated plots are shown. In these plots, the 68% and 95% CL are shown in orange and blue, respectively. In the same plot some other points are shown: the gray points represent the points out of the 2σ CL and black and green point represents the point which mixing angle variation surpass the 1% restriction. However, the green points, despite being out of the 1% restriction, the are still inside the 95% CL.

For BP1 and BP2, shown in Fig. 4.3, and BP3, shown in the *left* plot of Fig. 4.4, two different regions can be distinguished. The left region is located near the SM value for κ_λ , which tends to increase the total cross section w.r.t the best fit point in the selected benchmark plane, but at the same time, negative values of λ_{hhH} that are smaller than those at the best-fit point compensate for this increase. This is because, while the product of λ_{hhH} and Y_t^H tends to enhance the total cross-section, smaller values of λ_{hhH} neutralize this effect, as shown in Sec. 4.1. This effect finally tends to compensate the variation in the total cross section, giving a value closer to the selected benchmark point.

On the other hand, for BP4 in the *right* plot of Fig. 4.4, BP5 and BP6 in Fig. 4.5, and BP7 and BP8 in Fig. 4.6, the previous mentioned regions are not shown, resulting in a unique contour for all the generated points. This union of the two regions arises from the increased uncertainty in the total cross section. Consequently, This uncertainty yields smaller values for the χ^2 statistic, which widens the contours significantly compared to the first three benchmark points.

Due to the low uncertainty in $\Delta\sigma(pp \rightarrow hh)^{\text{BP}}$, the contours of the χ^2 fit were separated into two regions for the first three benchmark points: One for values of $\lambda_{hhH} > 0$ and the other for values

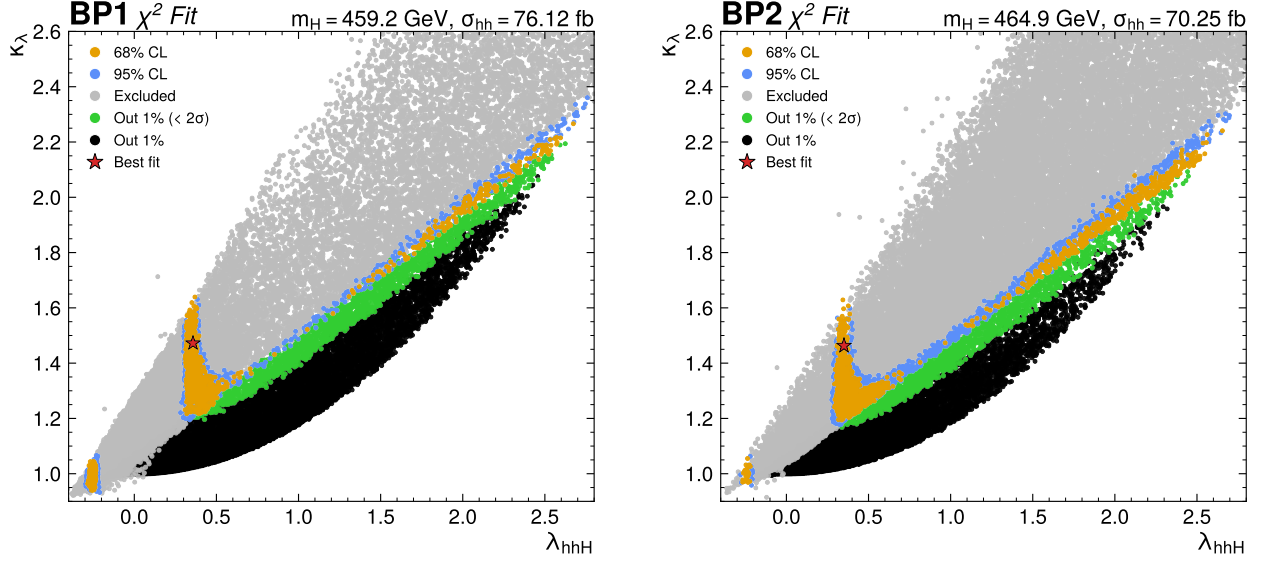


Figure 4.3: Generated points distribution on the $\kappa_\lambda - \lambda_{hhH}$ plane. (Left) point distribution of BP1, (right) points distribution of BP2.

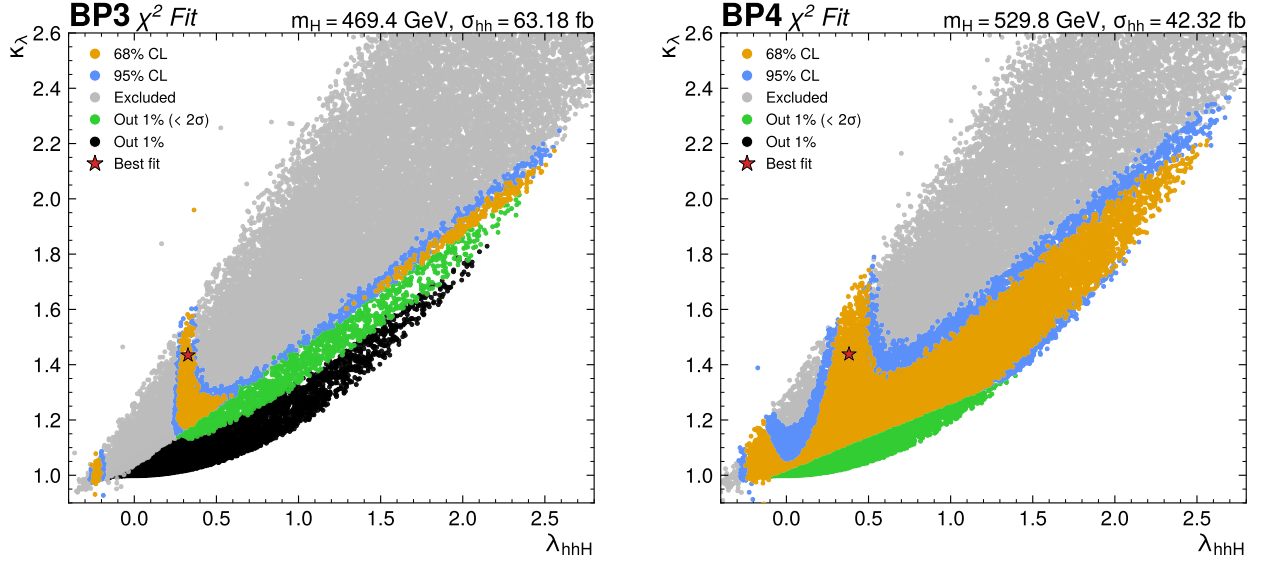


Figure 4.4: Generated points distribution on the $\kappa_\lambda - \lambda_{hhH}$ plane. (Left) point distribution of BP3, (right) points distribution of BP4.

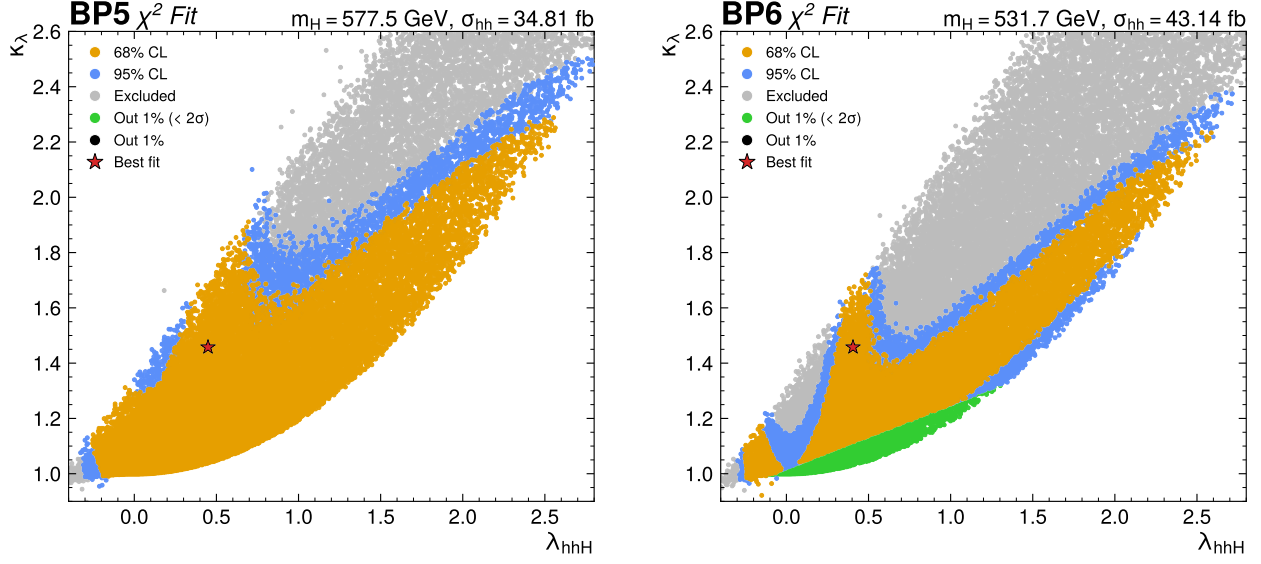


Figure 4.5: Generated points distribution on the $\kappa_\lambda - \lambda_{hhH}$ plane. (Left) point distribution of BP5, (right) points distribution of BP6.

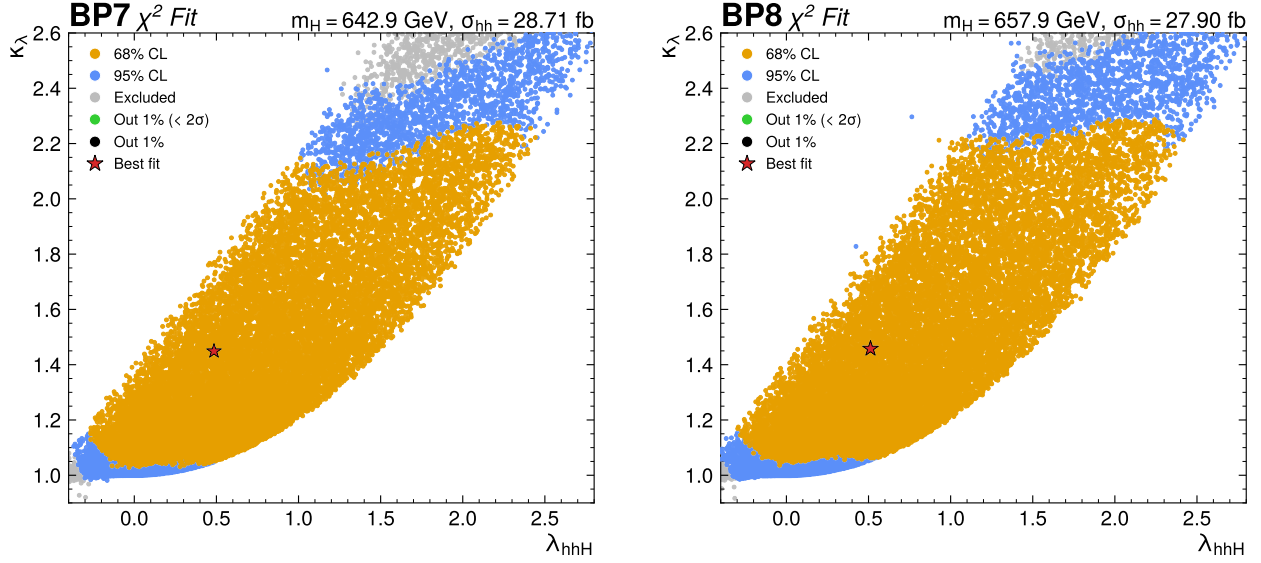


Figure 4.6: Generated points distribution on the $\kappa_\lambda - \lambda_{hhH}$ plane. (Left) point distribution of BP7, (right) points distribution of BP8.

of $\lambda_{hhH} < 0$. These two regions arise from a compensation between the κ_λ and λ_{hhH} couplings, as mentioned before. For values of κ_λ close to the SM, the interference of the light Higgs triangular and box diagrams occurs at lower values of the invariant mass, increasing the total cross section. However, a negative λ_{hhH} coupling yields a positive value for the $Y_t^H \times \lambda_{hhH}$ product, which tends to increase the total cross section, thus compensating for the decrease caused by the κ_λ effect. The same effect is presented for the rest of the benchmark points; however, their error in $\Delta\sigma$ is significantly higher, resulting in broader continuous χ^2 contours. In Tabs. 4.6 and 4.7, the upper and lower limits at 1σ and 2σ of each of the studied couplings for the first three benchmark points at the HL-LHC are presented. These bounds are shown for the two distinct regions obtained. In addition, in Tabs. 4.8 and 4.9, the upper and lower limits of the studied couplings for the rest of the benchmark points are shown.

BP	m_H [GeV]	$\kappa_\lambda^{\text{BP}}$	$\lambda_{hhH}^{\text{BP}}$	κ_λ ($\lambda_{hhH} < 0$)	κ_λ ($\lambda_{hhH} > 0$)	λ_{hhH} ($\lambda_{hhH} < 0$)	λ_{hhH} ($\lambda_{hhH} > 0$)
1	459.2	1.473	0.355	[0.9406 , 1.0452]	[1.2169 , 2.2622]	[-0.2854 , -0.2304]	[0.3097 , 2.6705]
2	464.9	1.463	0.326	[0.9530 , 1.0580]	[1.1876 , 2.2372]	[-0.2687 , -0.2186]	[0.2926 , 2.6544]
3	469.4	1.435	0.326	[0.9306 , 1.0777]	[1.1788 , 2.1210]	[-0.2543 , -0.1986]	[0.2688 , 2.5025]

Table 4.6: Upper and lower bounds of κ_λ and λ_{hhH} at 68% CL for the first three points. The bounds of each variable are separated for the regions where $\lambda_{hhH} > 0$ or $\lambda_{hhH} < 0$.

BP	m_H [GeV]	$\kappa_\lambda^{\text{BP}}$	$\lambda_{hhH}^{\text{BP}}$	κ_λ ($\lambda_{hhH} < 0$)	κ_λ ($\lambda_{hhH} > 0$)	λ_{hhH} ($\lambda_{hhH} < 0$)	λ_{hhH} ($\lambda_{hhH} > 0$)
1	459.2	1.473	0.355	[0.9339 , 1.0673]	[1.1962 , 2.3590]	[-0.2962 , -0.2110]	[0.2810 , 2.7541]
2	464.9	1.463	0.326	[0.9440 , 1.0588]	[1.1783 , 2.3154]	[-0.2778 , -0.2057]	[0.2712 , 2.7143]
3	469.4	1.435	0.326	[0.9280 , 1.0814]	[1.1517 , 2.2426]	[-0.2665 , -0.1803]	[0.2409 , 2.5851]

Table 4.7: Upper and lower bounds of κ_λ and λ_{hhH} at 95% CL for the first three points. The bounds of each variable are separated for the regions where $\lambda_{hhH} > 0$ or $\lambda_{hhH} < 0$.

BP	m_H [GeV]	$\kappa_\lambda^{\text{BP}}$	$\lambda_{hhH}^{\text{BP}}$	κ_λ	λ_{hhH}
4	529.8	1.439	0.380	[0.8968 , 2.2089]	[-0.2471 , 2.5609]
5	577.5	1.459	0.448	[0.9927 , 2.2846]	[-0.2494 , 2.5611]
6	531.7	1.457	0.404	[0.9237 , 2.2325]	[-0.2565 , 2.5702]
7	642.9	1.450	0.486	[1.0290 , 2.2758]	[-0.2610 , 2.4413]
8	657.9	1.458	0.511	[1.0384 , 2.2884]	[-0.2927 , 2.4153]

Table 4.8: Upper and lower limits of the studied variables for the rest of the benchmark points at 68% CL.

From the results obtain in Tabs. 4.6, 4.7, 4.8 and 4.9, we can conclude that depending which BP is realized in nature, a first determination of λ_{hhH} might be possible.

BP	m_H [GeV]	$\kappa_\lambda^{\text{BP}}$	$\lambda_{hhH}^{\text{BP}}$	κ_λ	λ_{hhH}
4	529.8	1.439	0.380	[0.8968 , 2.3628]	[-0.2844 , 2.6732]
5	577.5	1.459	0.448	[0.9775 , 2.4992]	[-0.3106 , 2.7862]
6	531.7	1.457	0.404	[0.9237 , 2.3754]	[-0.2938 , 2.7043]
7	642.9	1.450	0.486	[0.9824 , 2.6757]	[-0.3680 , 2.9226]
8	657.9	1.458	0.511	[0.9892 , 2.7138]	[-0.3917 , 2.8534]

Table 4.9: Upper and lower limits of the studied variables for the rest of the benchmark points at 95% CL.

The main goal of this master thesis is the investigation of the Higgs Singlet Extension of the SM. This model can accommodate a SFOEWPT, which fulfills the third Sakharov condition, one of the requirements necessary to explain the baryon asymmetry of the universe via electroweak baryogenesis, a mechanism that the SM cannot predict..

Based on [12], [16] a 2 dimension benchmark plane in the RxSM was analyzed, featuring a SFOEWPT. The phenomena playing in this Benchmark plane was investigated numerically. For further investigation eight benchmark points were chosen in this plane, taking into account the theoretical and experimental constrains of the model using **HiggsBounds** and **HiggsSignals**

As a next step, to validate the RxSM di-Higgs cross sections computed with **anyBSM** at the HL-LHC, the results were cross-checked against **HPAIR**. Initial discrepancies in the total cross sections led to an analysis of the invariant mass distributions for BP1 which revealed opposite differential structures: **anyBSM** predicted a dip-peak shape, whereas **HPAIR** predicted a peak-dip shape. The disagreement was traced to the sign of the product between the heavy Higgs top-Yukawa coupling and the trilinear coupling λ_{hhH} . Ultimately, an implementation inconsistency in the Y_t^H coupling was identified and successfully resolved.

Starting with the numerical analysis, the first part of it consisted on calculating the total cross section at tree level and using NLO in QCD precision for the three possible di-Higgs final states: hh, hH, HH in the gluon fusion channel, across the selected benchmark points from Tab. 2.1. With the calculations of the total cross section per each of the considered final states, the number of expected events were calculated using the most relevant final states at HL-LHC. These number of expected events were used to obtain the significance of each type of di-Higgs process for the different benchmark points. The results of the significances demonstrated that the events where at least one heavy Higgs is produced (hH and HH final states) were not sufficiently significant to observe them at the end of the HL-LHC phase, thus, only the double light Higgs production was studied, since the observation of the other processes were negligible.

In the second part of the analysis, a χ^2 fit was done. Since the processes with any heavy Higgs

boson were discarded due to their low significance, only the couplings entering in the $pp \rightarrow hh$ process were studied: λ_{hhH} and κ_λ . To extract their upper and lower limits, more than 35,000 points per benchmark point were generated under the imposed theoretical and experimental constraints. For the χ^2 fit, the total cross section $\sigma(pp \rightarrow hh)$ and κ_λ were considered and the extraction of the upper and lower bounds for each of the THCs via the minimization of the χ^2 equation (Eq. 4.1) was done. These bounds were obtained within a 68% 1σ and 95% (2σ) of confidence level.

The resulting contours revealed different structures among the evaluated benchmark points. For the first three benchmark points (BP1, BP2, BP3), the low uncertainty in the total cross section allowed two isolated regions to be resolved corresponding to $\lambda_{hhH} < 0$ and $\lambda_{hhH} > 0$. These two regions are presented for different values of κ_λ , one near the SM value and the other region for larger values of this parameter. When κ_λ approaches the SM value, the shifting interference pattern increases the total cross section, but a negative λ_{hhH} yields a positive $Y_t^H \times \lambda_{hhH}$ product that counterbalances this variation, yielding a total cross section close to the best fit value. Alternatively, for the remaining benchmark points, these two regions disappear and only one region is observed. This behavior is driven by the higher uncertainties in their total cross sections, which reduce the χ^2 statistic and widen the confidence intervals.

A

Appendix

A.1 Total cross section expressions for two scalar Higgs bosons

$$g^a g^b \rightarrow H_i H_j$$

In this section, the individual contributions of the studied Feymann diagrams that enter the calculation of the total and differential cross sections are studied. This derivation is based on the formalism presented in [41]. For the understanding, the construction of the matrix elements will be evaluated at LO, although the calculations done during this thesis are at NLO, since the results at NLO in QCD are the same but a factor ~ 2 larger [42].

As mentioned in Chap. 3, all the processes studied in this work are generated via gluon-gluon fusion to produce a pair of scalar Higgs boson through different interactions. Since gluons are QCD bosons, they carry one of eight possible color states. However, the final states are colorless, thus the initial gluons must carry exactly the opposite color charges to conserve the color.

Taking into account all the diagrams discussed in Chap. 3, the element matrix for the production of two scalar Higgs bosons can be written as:

$$\mathcal{M}(g^a g^b \rightarrow H_i H_j) = \frac{G_F \alpha_s \hat{s}}{2\sqrt{2}\pi} \mathcal{A}_{\mu\nu} \epsilon_a^\mu \epsilon_b^\nu \delta_{ab} \ , \quad (\text{A.1})$$

where $a, b \in \{1, \dots, 8\}$ represent the gluon color indices, H_i, H_j denote the scalar bosons, and G_F is the Fermi constant. The $\epsilon_{i\mu}$ are the polarization vectors of the incoming gluons, while $\mathcal{A}^{\mu\nu}$ is the tensor that encapsulates the full dynamic and kinematic structure of the diagrams, which is decomposed as:

$$\mathcal{A}_{\mu\nu} = F_1 T_{1\mu\nu} + F_2 T_{2\mu\nu} \ . \quad (\text{A.2})$$

[41]

This decomposition of the tensor isolates the purely kinematic tensor projectors $T_i^{\mu\nu}$ from the scalar form factors.

This decomposition isolates the purely kinematic tensor projectors, $T_i^{\mu\nu}$, from the scalar form factors, F_i , which contain the underlying quantum dynamics. The orthogonal tensor structures $T_1^{\mu\nu}$ and $T_2^{\mu\nu}$ are defined as:

$$T_1^{\mu\nu} = g^{\mu\nu} - \frac{p_a^\nu p_b^\mu}{(p_a p_b)} , \quad (\text{A.3})$$

$$T_2^{\mu\nu} = g^{\mu\nu} + \frac{p_c^2 p_a^\nu p_b^\mu}{p_T^2 (p_a p_b)} - \frac{2(p_b p_c) p_a^\nu p_c^\mu}{p_T^2 (p_a p_b)} - \frac{2(p_a p_c) p_b^\mu p_c^\nu}{p_T^2 (p_a p_b)} + \frac{2p_c^\mu p_c^\nu}{p_T^2} , \quad (\text{A.4})$$

where p_a and p_b denote the four-momenta of the two incoming gluons, p_c represents the four-momentum of one of the outgoing Higgs bosons and the transverse momentum squared is given by $p_T^2 = \frac{2(p_a p_c)(p_b p_c)}{(p_a p_b)} - p_c^2$.

While, the form factors, which indicates how the interferences between diagrams occurs, are written as following:

$$F_1 = C_\Delta^h F_\Delta + C_\Delta^H F_\Delta + C_\square F_\square, \quad F_2 = C_\square G_\square . \quad (\text{A.5})$$

where Δ represents the triangle diagram of the interaction and $\square$ represents the box diagram. Notice that F_1 receives contributions from both the s -channel triangle diagrams (F_Δ) and the box diagrams ($F_\square$), while F_2 is generated exclusively by the box topologies. From the expression for F_1 , we can explicitly isolate the couplings $C_\Delta^{H_k}$ and $C_\square$ associated with the scalar bosons:

$$C_\Delta^{H_k} = \frac{\lambda_{H_i H_j H_k} Y_t^{H_k} v}{\hat{s} - m_{H_k}^2 + i m_{H_k} \Gamma_{H_k}} , \quad C_\square = Y_t^{H_i} Y_t^{H_j} , \quad (\text{A.6})$$

where $\lambda_{H_i H_j H_k}$ is the THCs, $\hat{s}$ is the partonic Mandelstam variable or the square of the center-of-mass energy and $Y_t^{H_k}$ is the scalar top-quark Yukawa. In the RxSM, $Y_t^{H_k}$ is modified due to the introduction of a new scalar bosons and the couplings are given by:

$$Y_t^h = y_t^{\text{SM}} \cos \alpha , \quad Y_t^H = -y_t^{\text{SM}} \sin \alpha , \quad (\text{A.7})$$

the Standard Model top Yukawa coupling is $y_t^{\text{SM}} = \sqrt{2} m_t / v$.

A.1.1 Double light Higgs boson production ($gg \rightarrow hh$)

After observing the results shown in Sec. 4.1.2 from the results chapter, the processes where any BSM Higgs bosons is processed, i.e. hH and HH processes, were omitted and the double light Higgs

production final state was study.

Applying the general formalism shown in the previous section and obtained from [41], the specific case of double light Higgs production ($H_i = h$ and $H_j = h$) was analyzed. For this process, the matrix element is composed of three distinct topological contributions (Fig. 3.1): the light Higgs s -channel mediation, the heavy Higgs s -channel mediation, and the box diagrams. Therefore, the couplings that contribute to $\mathcal{M}$ are:

$$C_{\Delta}^h = \frac{\lambda_{hhh} Y_t^h v}{\hat{s} - m_h^2 + im_h \Gamma_h}, \quad C_{\Delta}^H = \frac{\lambda_{hhH} Y_t^H v}{\hat{s} - m_H^2 + im_H \Gamma_H}, \quad C_{\square} = Y_t^h Y_t^h, \quad (\text{A.8})$$

leading to the matrix elements for each of the Feymann diagrams:

$$\mathcal{M}_{\Delta}^h = \frac{G_F \alpha_s \hat{s}}{2\sqrt{2}\pi} \left[\frac{\lambda_{hhh} Y_t^h v}{\hat{s} - m_h^2 + im_h \Gamma_h} \right] F_{\Delta} T_{1\mu\nu} \epsilon_a^{\mu} \epsilon_b^{\nu} \delta_{ab}, \quad (\text{A.9})$$

$$\mathcal{M}_{\Delta}^H = \frac{G_F \alpha_s \hat{s}}{2\sqrt{2}\pi} \left[\frac{\lambda_{hhH} Y_t^H v}{\hat{s} - m_H^2 + im_H \Gamma_H} \right] F_{\Delta} T_{1\mu\nu} \epsilon_a^{\mu} \epsilon_b^{\nu} \delta_{ab}, \quad (\text{A.10})$$

$$\mathcal{M}_{\square} = \frac{G_F \alpha_s \hat{s}}{2\sqrt{2}\pi} Y_t^h Y_t^h (F_{\square} T_{1\mu\nu} + G_{\square} T_{2\mu\nu}) \epsilon_a^{\mu} \epsilon_b^{\nu} \delta_{ab}. \quad (\text{A.11})$$

Summing all contributions and grouping the terms by their dependence on the orthogonal projectors $T_1^{\mu\nu}$ and $T_2^{\mu\nu}$, the full partonic amplitude is going by,

$$\begin{aligned} \mathcal{M}(g^a g^b \rightarrow hh) &= \frac{G_F \alpha_s \hat{s}}{2\sqrt{2}\pi} \delta_{ab} \epsilon_1^{\mu} \epsilon_2^{\nu} \\ &\times \left\{ T_{1\mu\nu} \left[F_{\Delta} \left(\frac{\lambda_{hhh} Y_t^h v}{\hat{s} - m_h^2 + im_h \Gamma_h} - \frac{\lambda_{hhH} Y_t^H v}{\hat{s} - m_H^2 + im_H \Gamma_H} \right) + F_{\square} (Y_t^h)^2 \right] \right. \\ &\left. + T_{2\mu\nu} G_{\square} (Y_t^h)^2 \right\}. \end{aligned} \quad (\text{A.12})$$

Finally, to express $\mathcal{M}$ entirely in terms of the fundamental parameters of the RxSM, we substitute the modified top-quark Yukawa couplings $Y_t^h = y_t^{\text{SM}} \cos \alpha$ and $Y_t^H = -y_t^{\text{SM}} \sin \alpha$. The fully expanded analytical expression for the double light Higgs production is given then by:

$$\begin{aligned} \mathcal{M}(g^a g^b \rightarrow hh) &= \frac{G_F \alpha_s \hat{s}}{2\sqrt{2}\pi} \delta_{ab} \epsilon_a^{\mu} \epsilon_b^{\nu} \\ &\times \left\{ T_{1\mu\nu} \left[F_{\Delta} \left(\frac{\lambda_{hhh} v y_t^{\text{SM}} \cos \alpha}{\hat{s} - m_h^2 + im_h \Gamma_h} - \frac{\lambda_{hhH} v y_t^{\text{SM}} \sin \alpha}{\hat{s} - m_H^2 + im_H \Gamma_H} \right) + F_{\square} (y_t^{\text{SM}} \cos \alpha)^2 \right] \right. \\ &\left. + T_{2\mu\nu} \left[G_{\square} (y_t^{\text{SM}} \cos \alpha)^2 \right] \right\}. \end{aligned} \quad (\text{A.13})$$

A.2 Matrix element behavior

When analyzing the χ^2 contours for BP1 (see Fig. 4.3), an interesting geometric distribution emerges if the sampled region is not strictly restricted to the 1% variation limit of the mixing angle α . To understand the underlying physics shaping these contours, three additional points from the BP1 parameter space were selected and compared. In Fig. A.1, the unconstrained distribution for BP1 is shown, where black markers indicate points outside the 95% CL ($\Delta\chi^2 > 5.99$), while gradient-colored markers represent allowed points within the 95% CL.

For a fixed $\kappa_\lambda = 1.5$, three points with varying λ_{hhH} values were chosen to compare their cross-section behavior against BP1. Based on their spatial location in the parameter distribution, these points are labeled LEFT (*red*), MIDDLE (*green*), and RIGHT (*orange*).

Specifically, the LEFT point represents a configuration with a slightly larger λ_{hhH} than BP1, leading to a moderate increase in the heavy Higgs width (Γ_H) and the cross section. The MIDDLE point is characterized by a significantly higher λ_{hhH} , which maximizes the resonant cross section but returns a heavy χ^2 penalty, pushing it out of the statistically favored region. Finally, the RIGHT point features the largest λ_{hhH} and decay width; however, its cross section lowers to values comparable to BP1, restoring a competitive χ^2 score. The Lagrangian parameters defining these points are detailed in Tab. A.1, and the corresponding physical observables computed with `anyBSM` are compared in Tab. A.2.

Point	$\cos \alpha$	m_H [GeV]	v_S [GeV]	κ_S [GeV]	κ_{SH} [GeV]
BP1	0.9840	459.20	46.30	-528.30	-345.57
LEFT	0.9806	459.25	166.54	-504.76	-681.83
MIDDLE	0.9925	459.19	480.78	-2381.55	-4124.72
RIGHT	0.9946	459.27	189.96	-7432.02	-2228.99

Table A.1: Lagrangian parameters of the selected points from the BP1 parameter space.

Point	λ_{hhH}	Γ_H [GeV]	σ_{hh} [fb]	σ_{hh}^{res} [fb]	χ^2
BP1	0.3550	2.0111	76.12	61.78	0.0
LEFT	0.3731	2.3453	81.62	66.32	3.2
MIDDLE	0.8451	3.8152	95.56	81.06	32.5
RIGHT	1.0820	5.6332	78.68	64.81	0.4

Table A.2: Physical observables and statistical performance for the selected points. Note that σ_{hh}^{res} is the total cross section driven exclusively by the resonant triangle diagram.

Because the χ^2 values are predominantly driven by deviations in the total cross section σ_{hh} , analyzing points with the same κ_λ allows us to isolate how the cross section evolves strictly as a function of the trilinear coupling λ_{hhH} . To understand the origin of these cross-section variations,

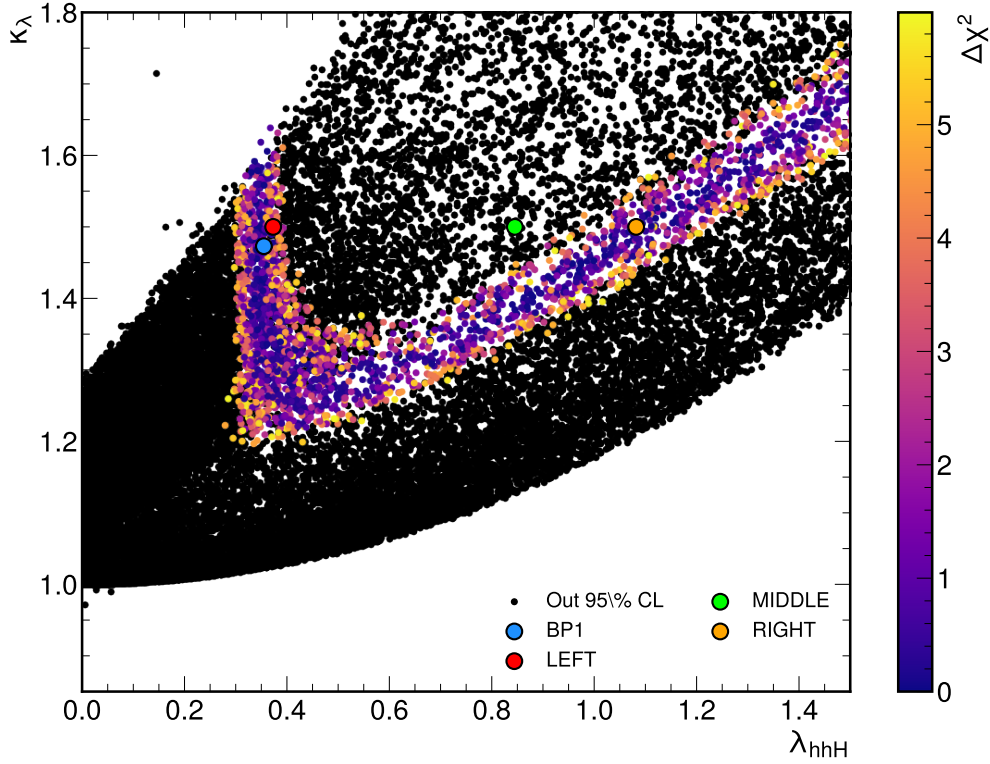


Figure A.1: Generated point distribution on the $\kappa_\lambda - \lambda_{hhH}$ plane without the restrictive mixing angle constraint. Black points indicate parameter sets excluded at 95% CL, while gradient-colored points fall within the allowed 95% CL boundary.

an analysis of the differential cross sections with respect to the invariant mass m_{hh} was performed.

When including all contributing diagrams (both triangle and box topologies), the invariant mass distributions differ primarily around the resonance peak, which is driven by the heavy Higgs s -channel mediation. As shown in Fig. A.2, the main differences manifest in the height and width of the resonance peak. Slight discrepancies are also visible in the low-mass region near $m_{hh} \approx 280$ GeV; these arise because the κ_λ values are not perfectly identical across the sample, despite all being close to $\kappa_\lambda \approx 1.5$.

Since the most prominent kinematic differences occur at the resonant peak, we isolate the contribution of the heavy Higgs triangle diagram. As observed in Tab. A.2, the total cross section σ_{hh} closely follows the underlying trend of the purely resonant cross section σ_{hh}^{res} . Figure A.3 isolates the invariant mass distribution for this resonant diagram, clearly illustrating the significant variations in both the peak height and the resonance width across the selected points.

These kinematic variations arise from two competing physical effects driven by the increase of the trilinear coupling λ_{hhH} . On one hand, a larger λ_{hhH} coupling increases the total decay width of the heavy Higgs boson (Γ_H), which inherently broadens the resonance structure. On other hand, this broadening effect effectively suppresses the maximum height of the resonance peak.

The delicate interplay between these two competing phenomena dictates the total integrated cross section: as λ_{hhH} increases, the integral initially grows, reaching a local maximum at an intermediate

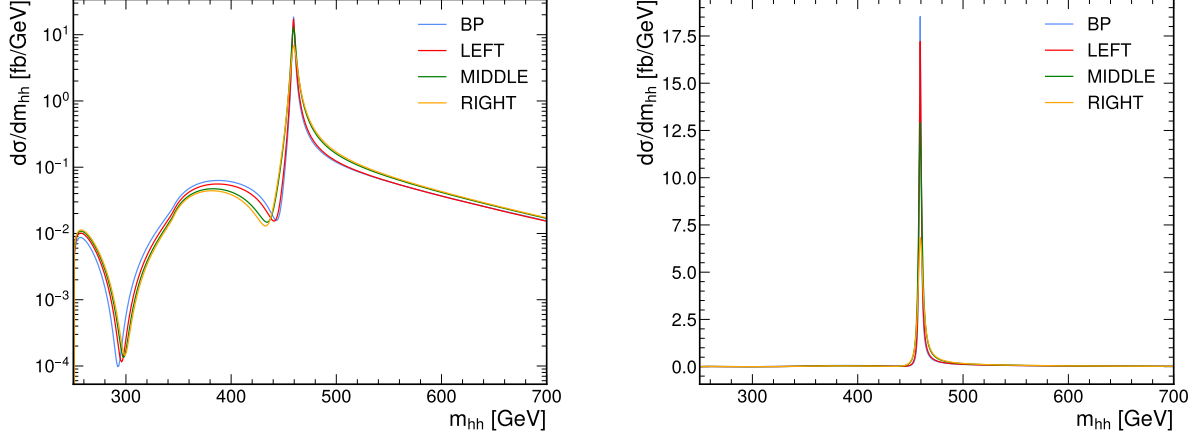


Figure A.2: Invariant mass distribution of the differential cross section including all diagrams (triangle and box) for the four selected points, displayed in (a) logarithmic scale and (b) linear scale.

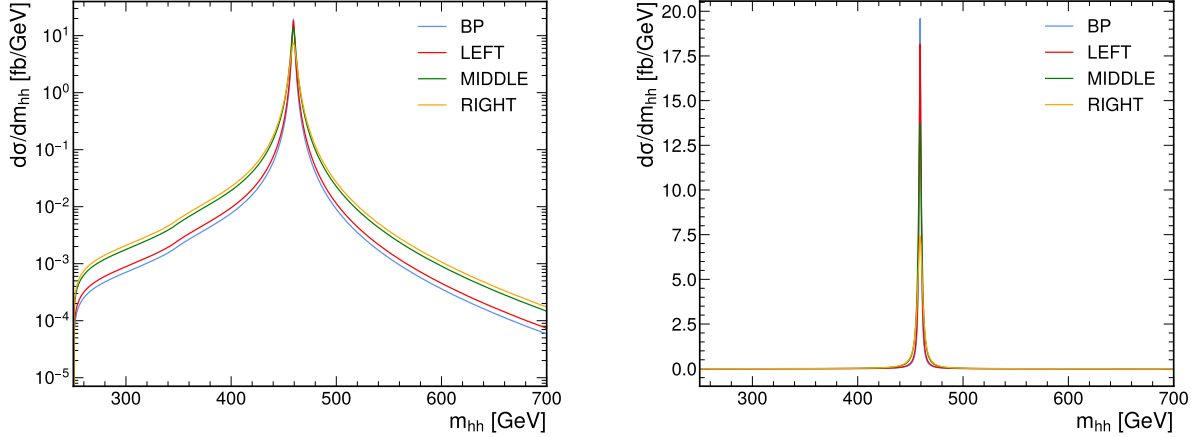


Figure A.3: Invariant mass distribution of the differential cross section considering exclusively the resonant diagram for the four selected points in (a) logarithmic scale and (b) linear scale.

value (represented by the MIDDLE point), before the peak suppression dominates and yields smaller total cross sections for points with even larger couplings. Because an exact analytical integration over this highly non-linear parameter space is not trivially obtained, a deeper mathematical expansion of the matrix element is omitted. Nevertheless, the non-trivial shape of the confidence contours is fundamentally explained by the balance of these two competing physical effects near the resonance.



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